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(54) **NEAR-FIELD AUDIO RENDERING**

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(2013.01); **H04R 5/04** (2013.01); **H04S 3/008**

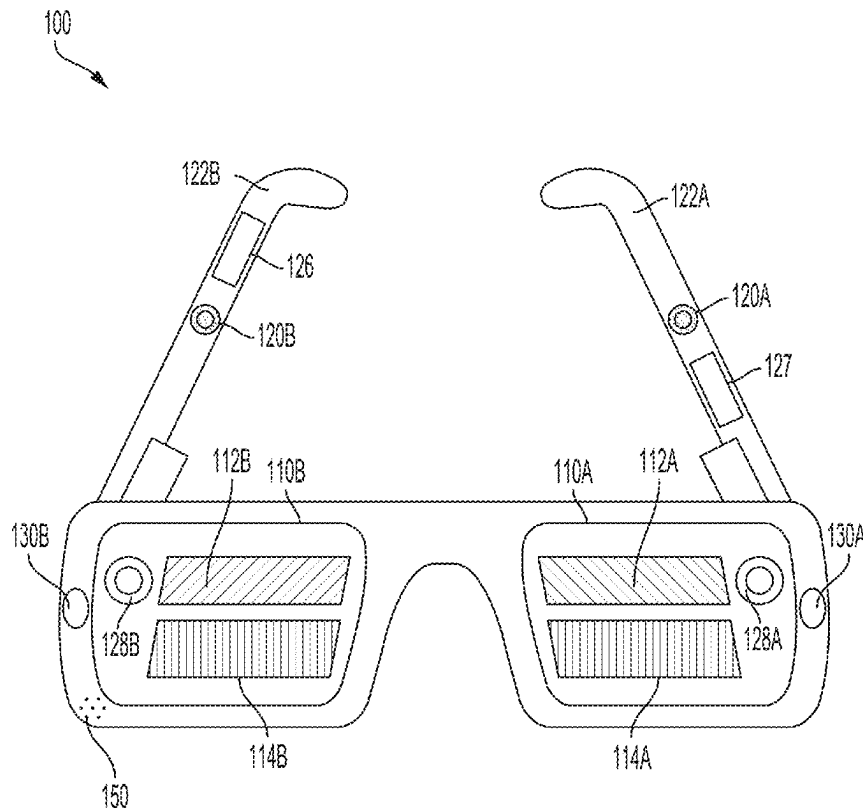
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2400/11 (2013.01); **H04S 2420/01** (2013.01)

(57)

ABSTRACT

Examples of the disclosure describe systems and methods for presenting an audio signal to a user of a wearable head device. According to an example method, a source location corresponding to the audio signal is identified. For each of the respective left and right ear of the user, a virtual speaker position, of a virtual speaker array, is determined, the virtual speaker position collinear with the source location and with a position of the respective ear. For each of the respective left and right ear of the user, a head-related transfer function (HRTF) corresponding to the virtual speaker position and to the respective ear is determined; and the output audio signal is presented to the respective ear of the user via one or more speakers associated with the wearable head device. Processing the audio signal includes applying the HRTF to the audio signal.



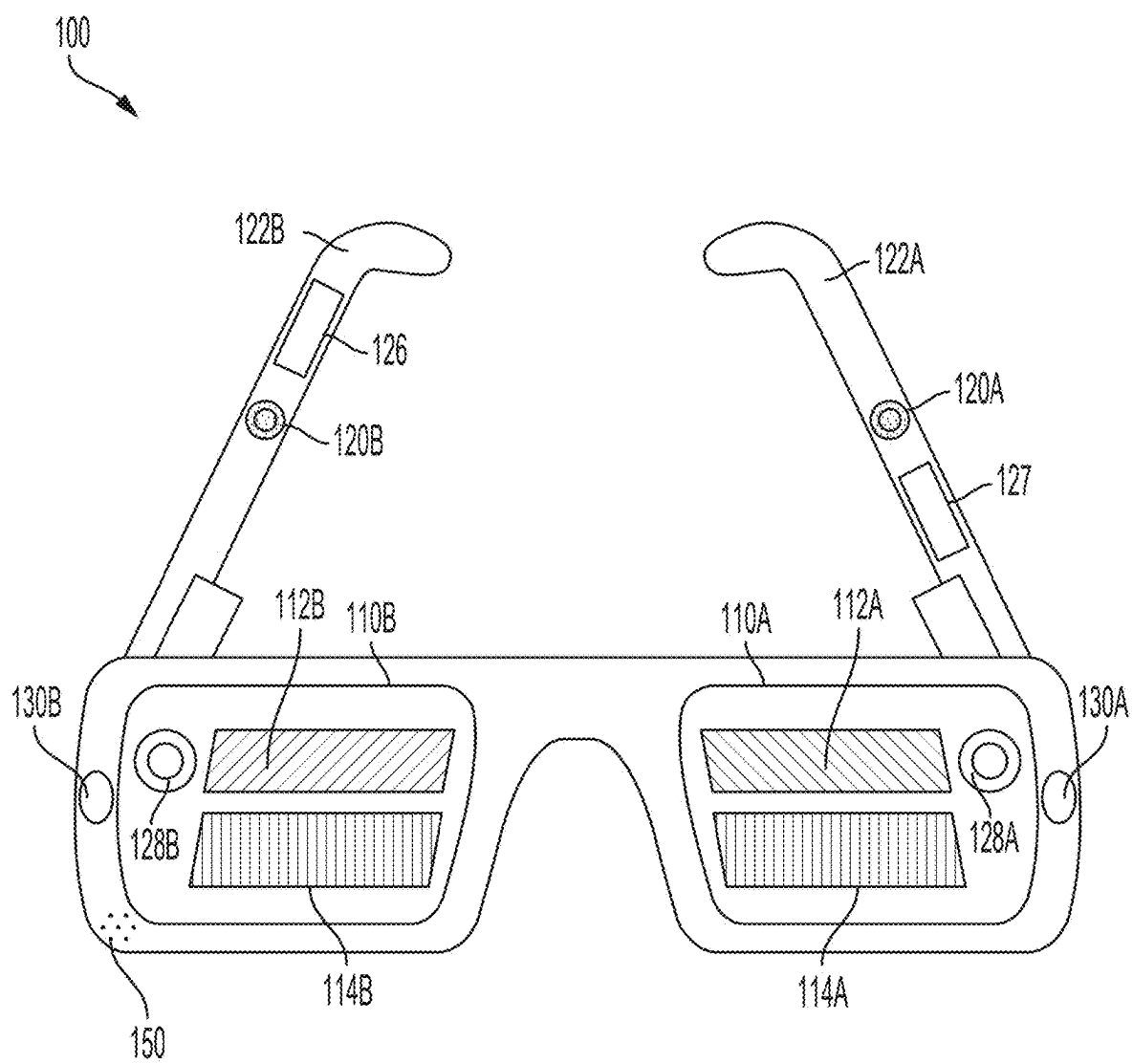


FIG. 1

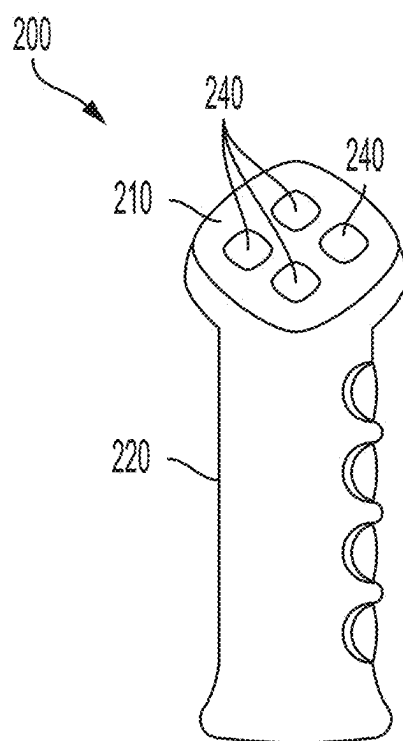


FIG. 2

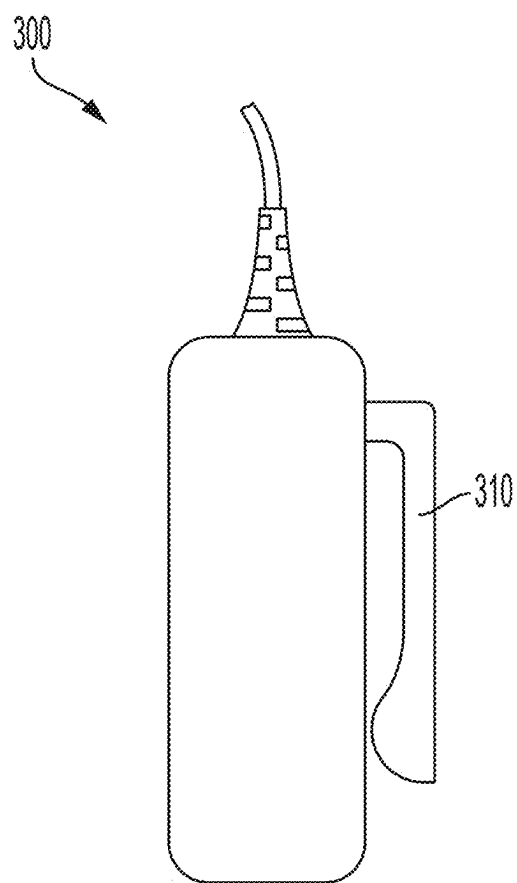


FIG. 3

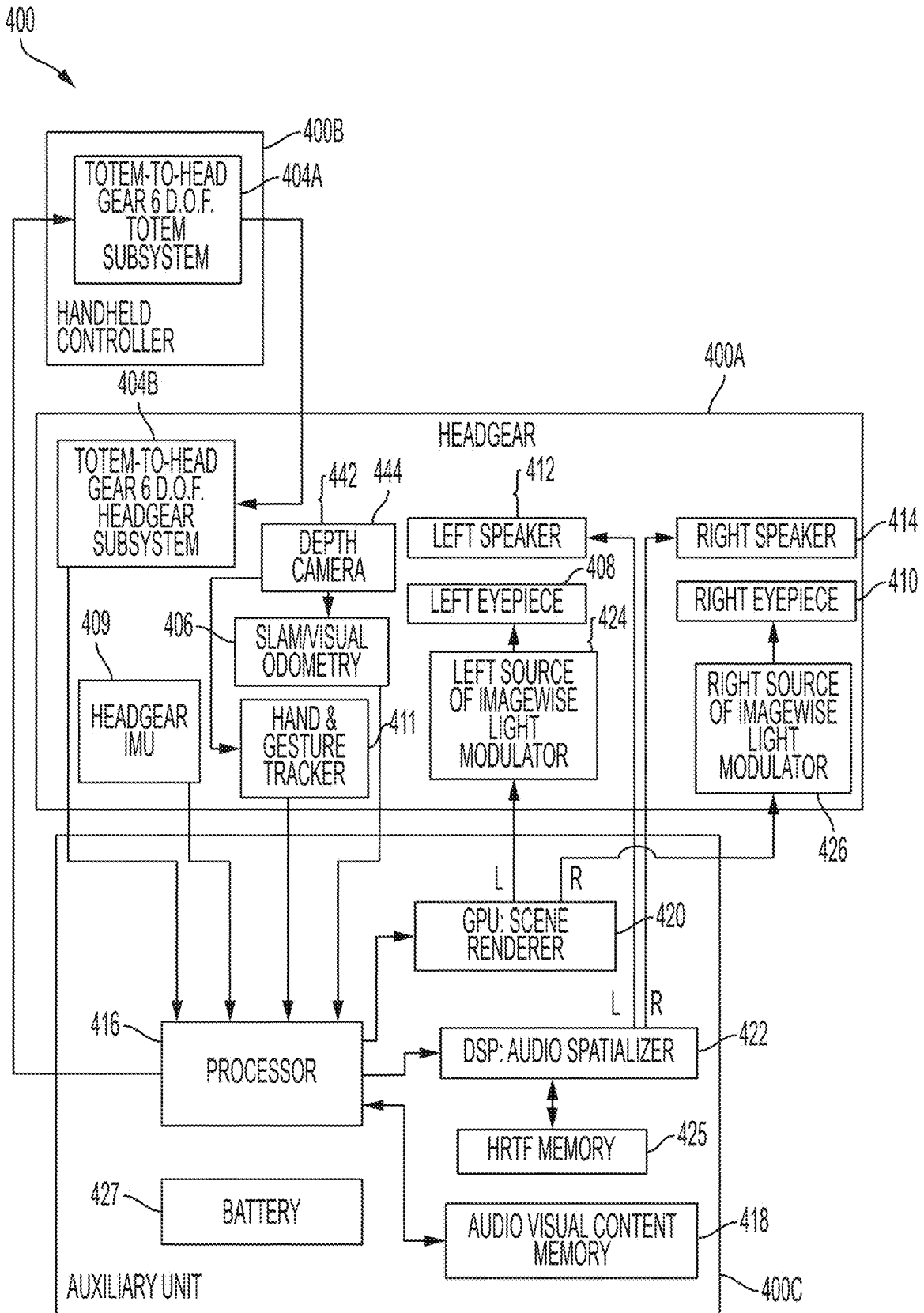


FIG. 4

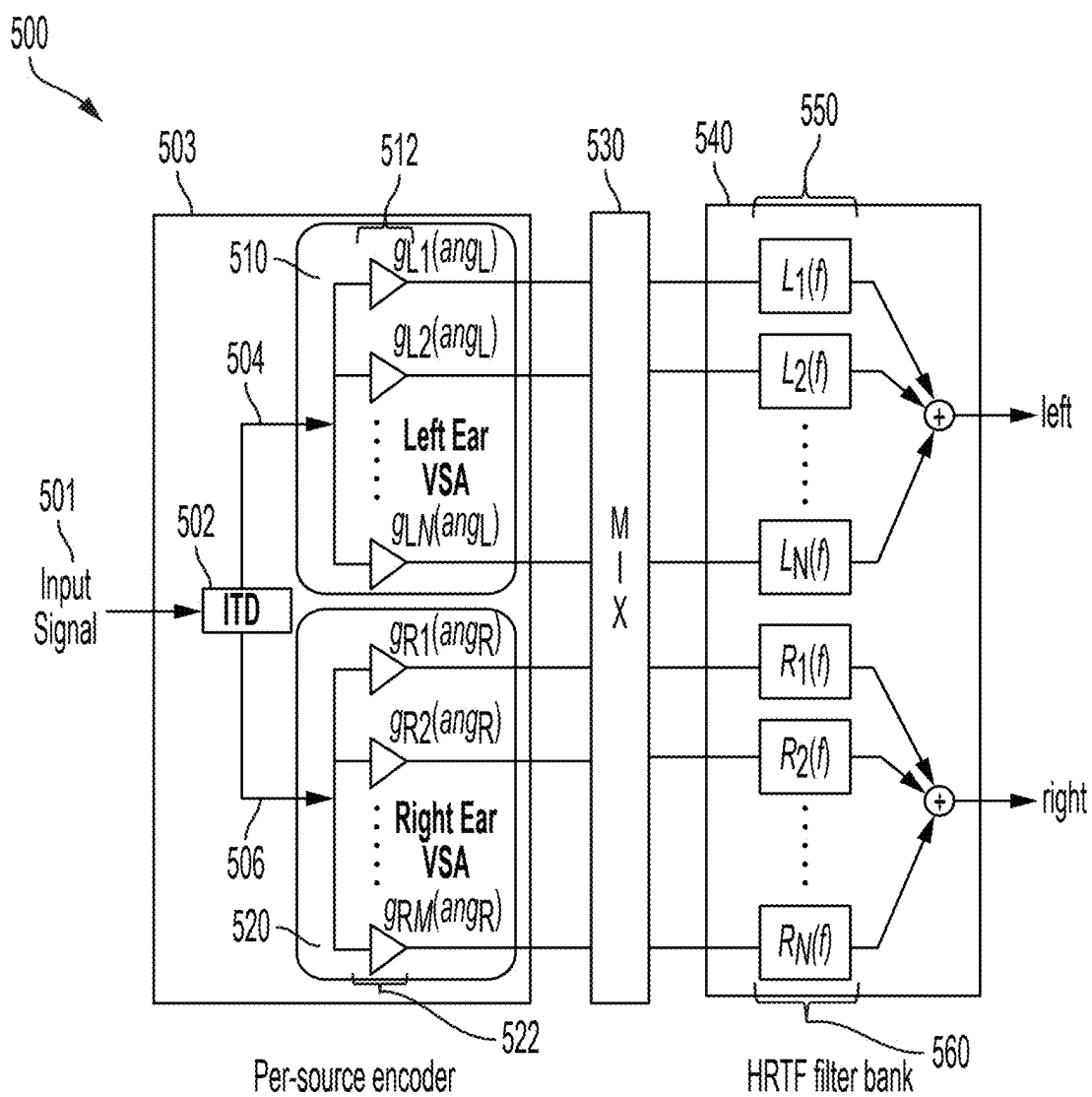


FIG. 5

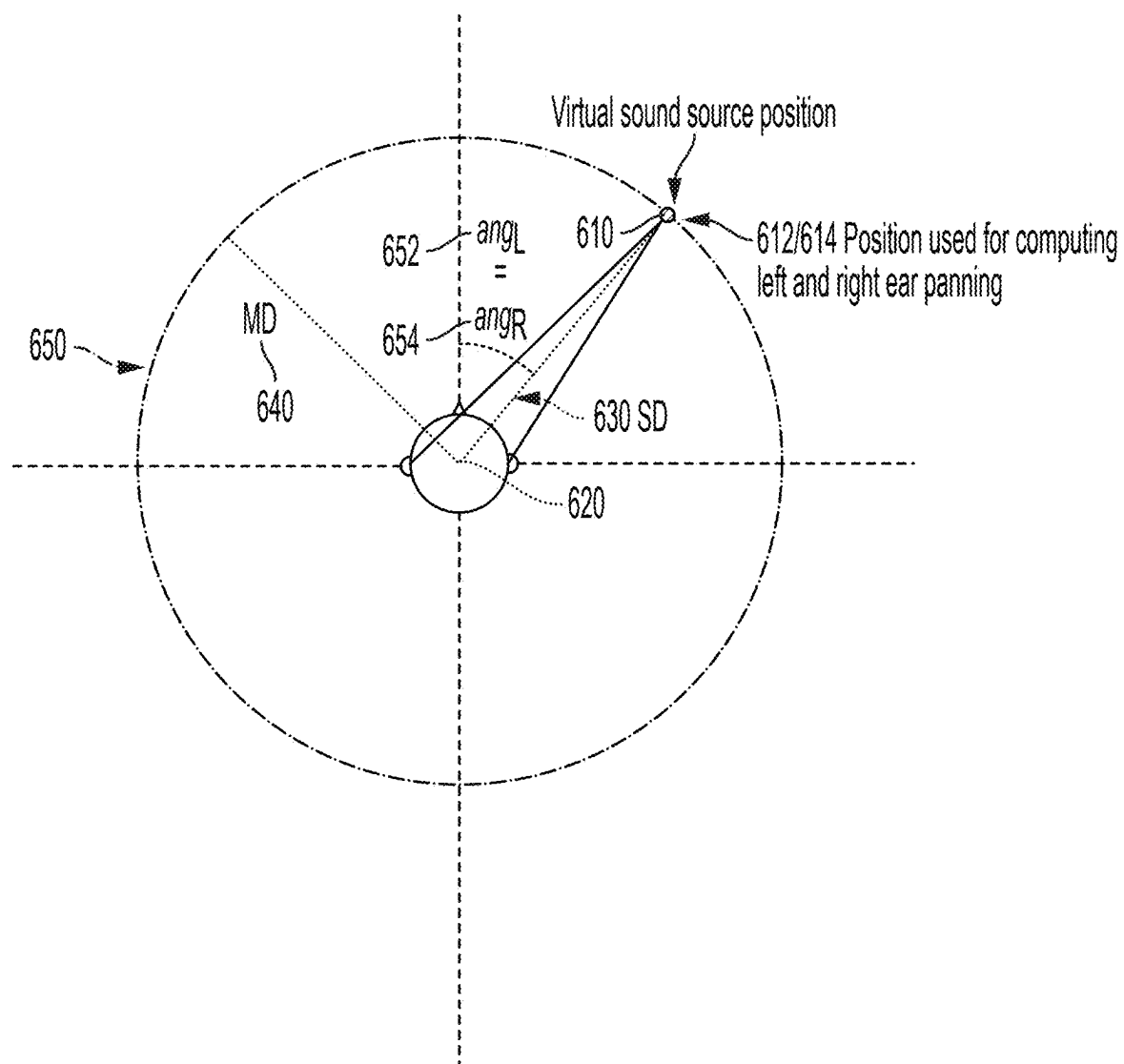


FIG. 6A

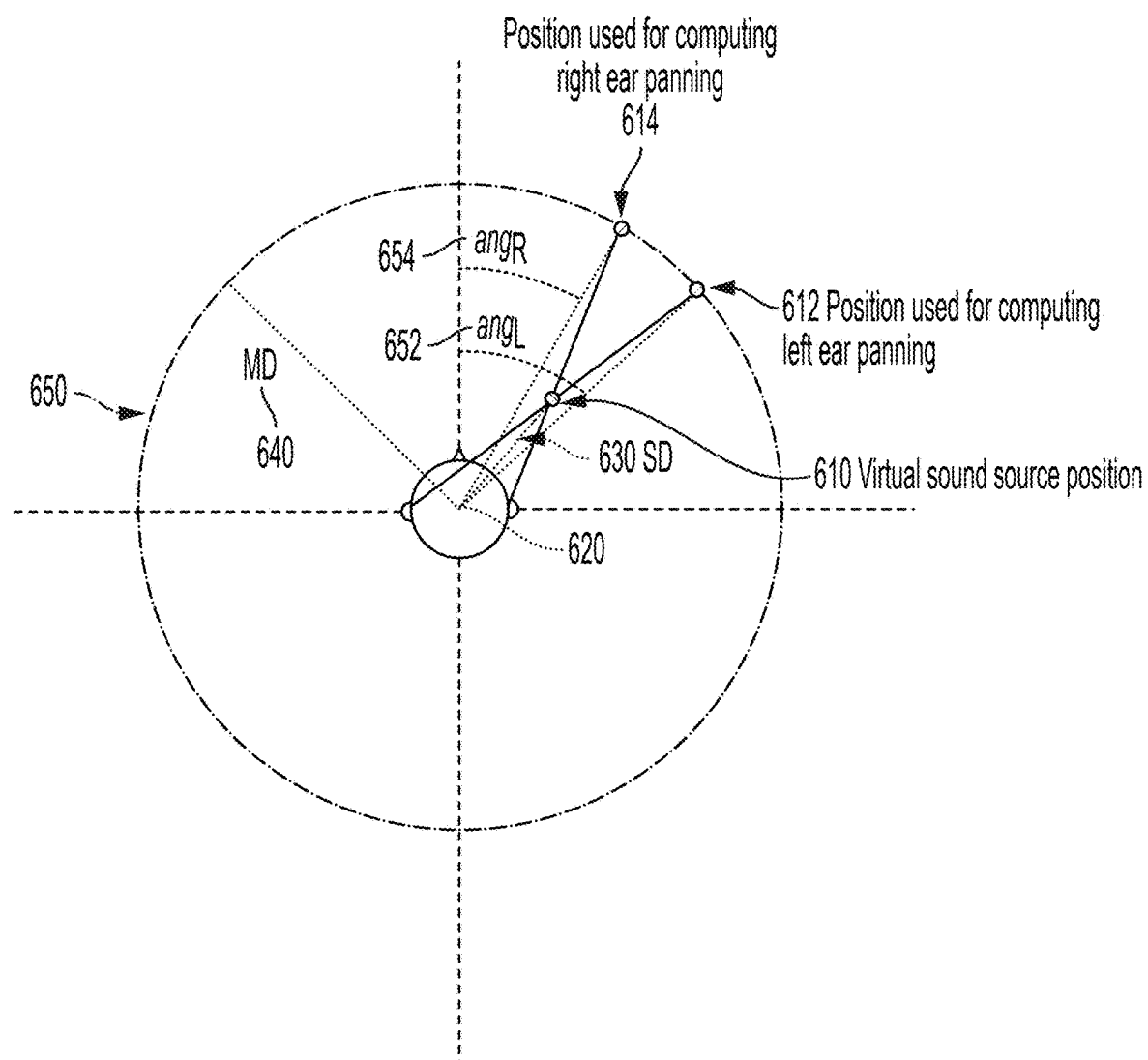


FIG. 6B

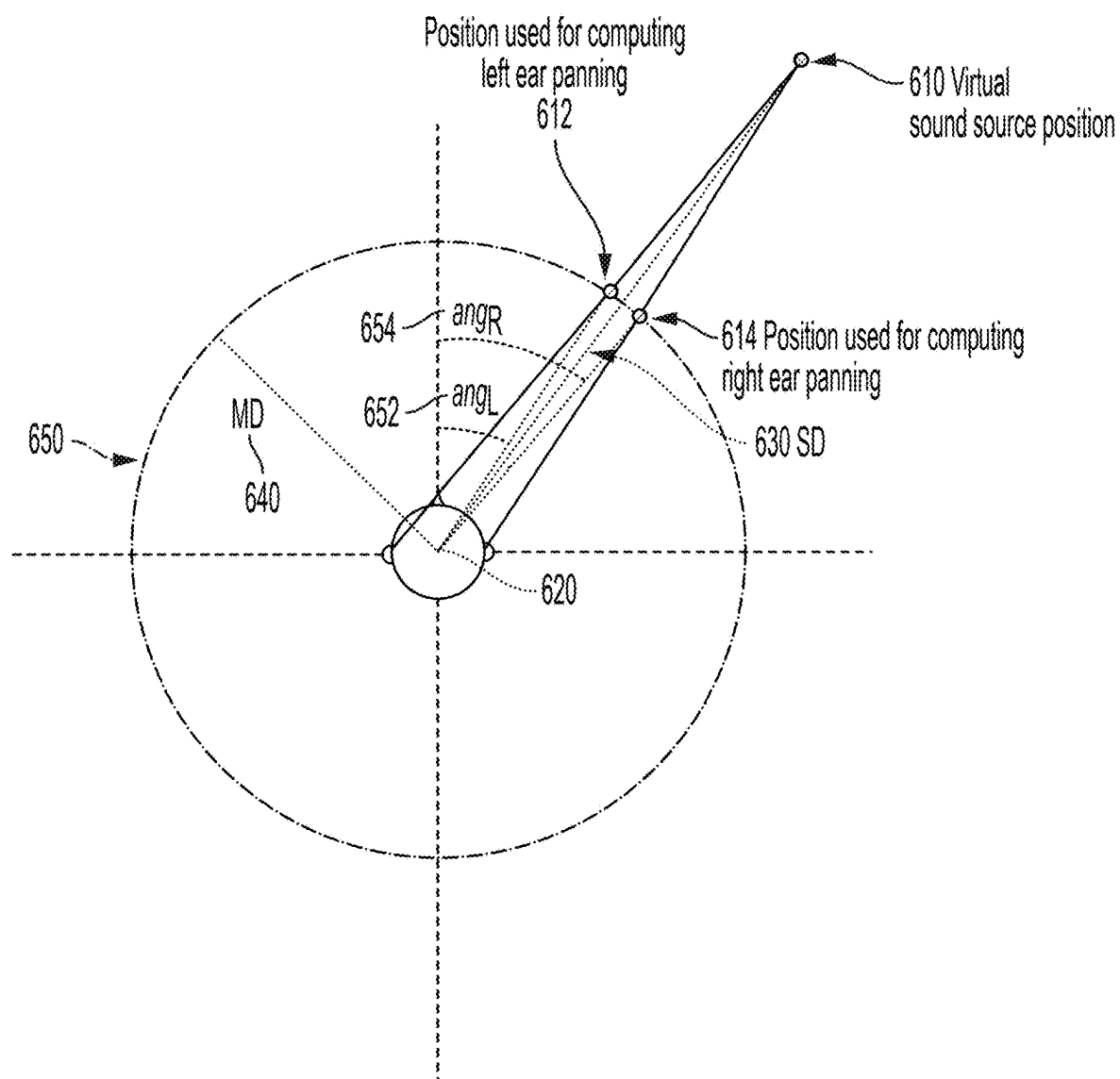


FIG. 6C

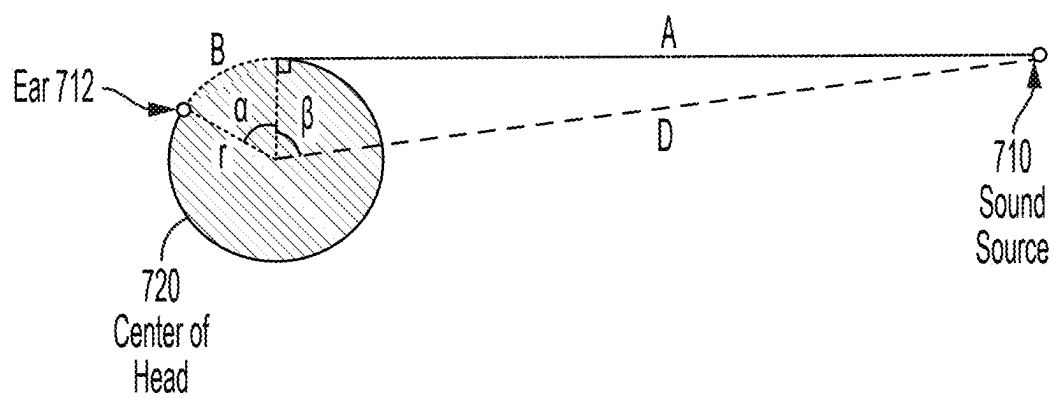
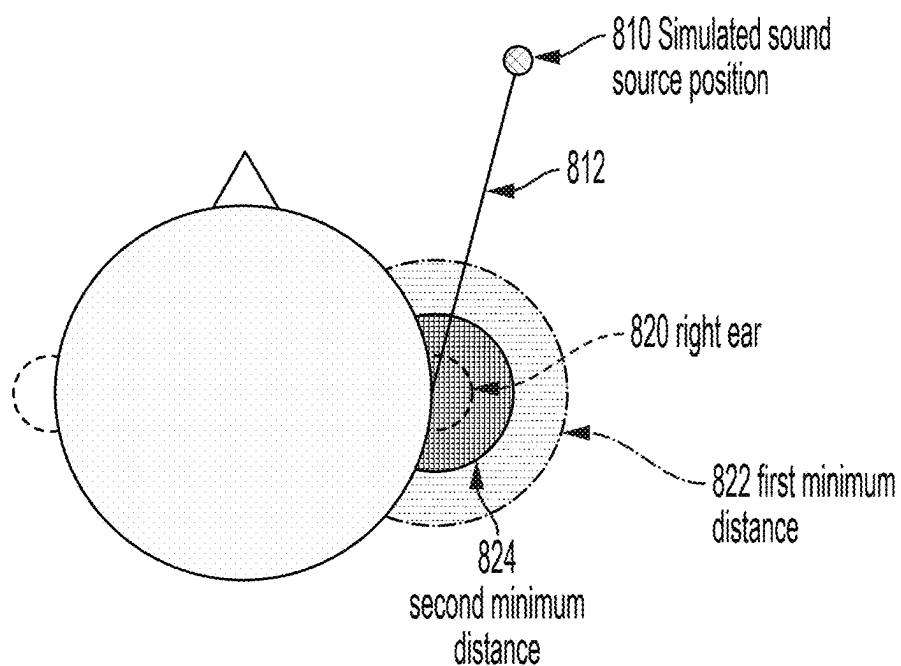
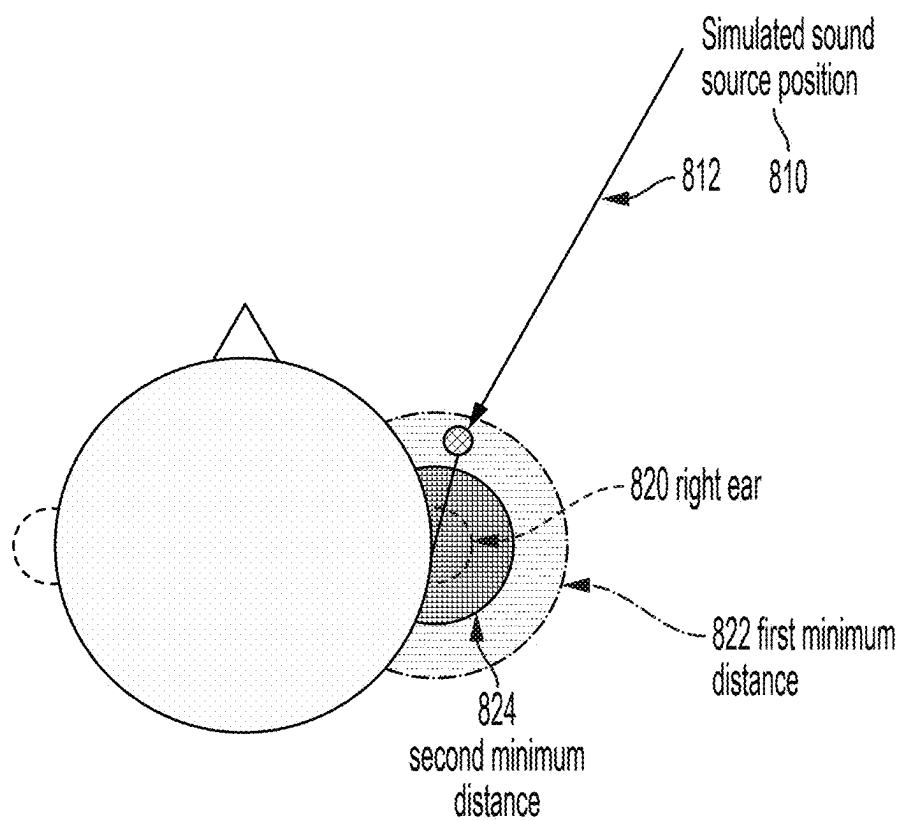


FIG. 7



-  Zone where object distance for all computation is clamped to the second minimum distance
-  Zone where object distance for all gain computation is clamped to the first minimum distance

FIG. 8A



-  Zone where object distance for all computation is clamped to the second minimum distance
-  Zone where object distance for all gain computation is clamped to the first minimum distance

FIG. 8B

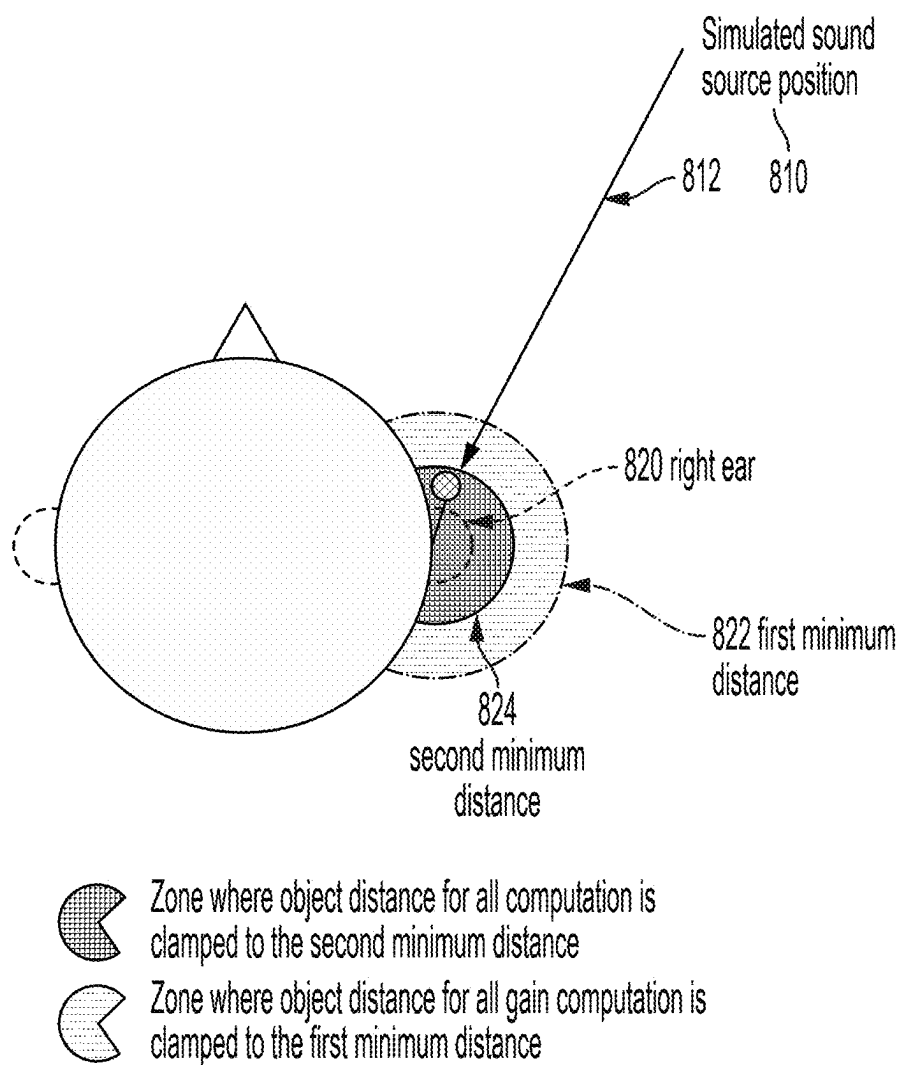


FIG. 8C

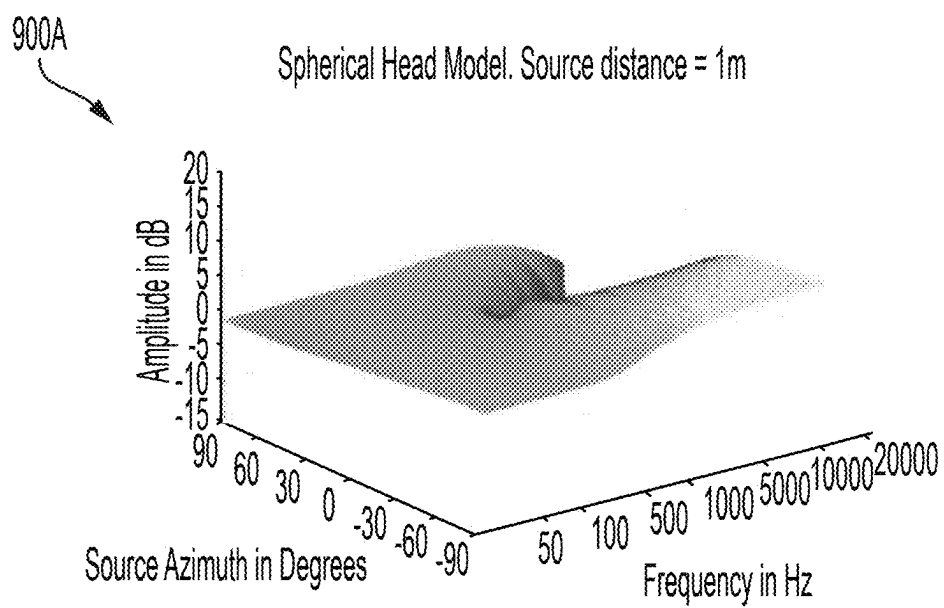


FIG. 9A

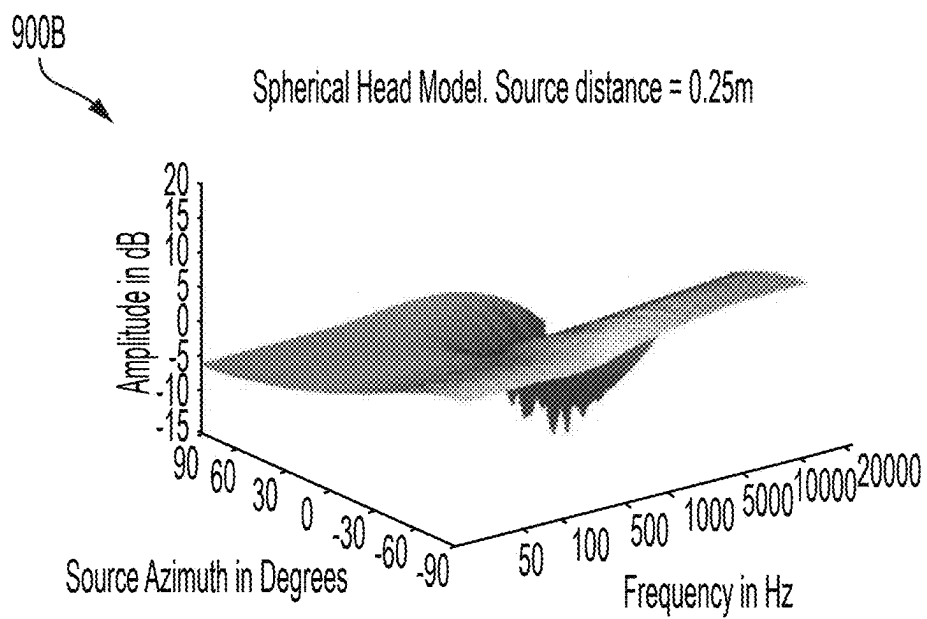


FIG. 9B

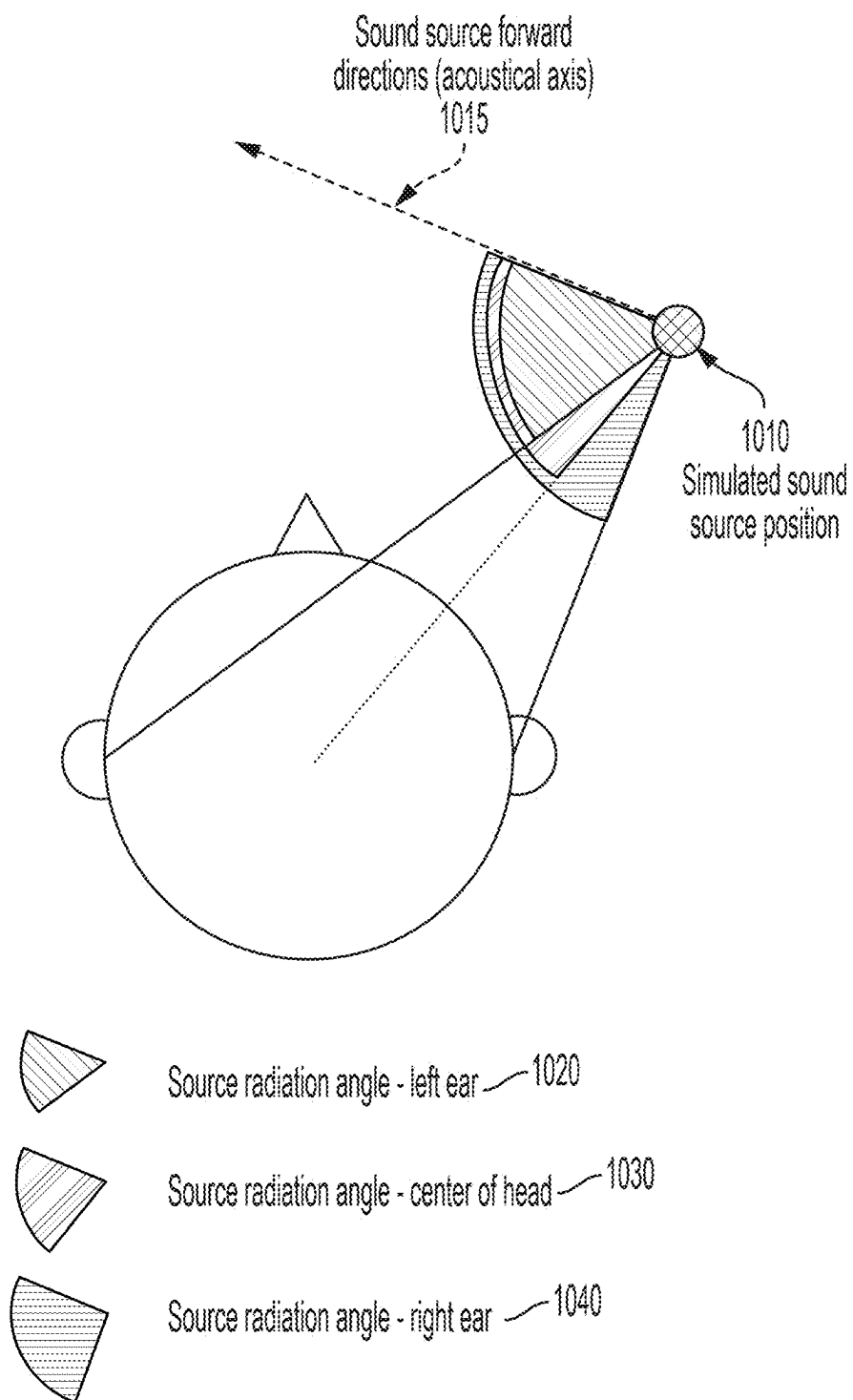


FIG. 10

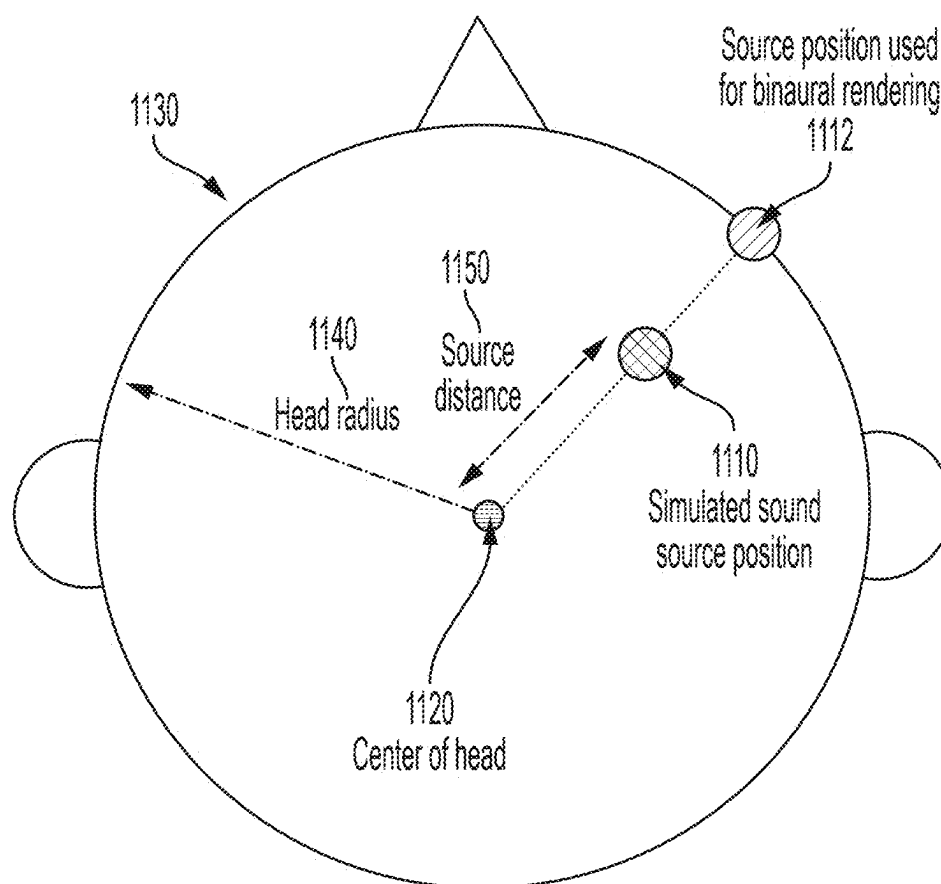


FIG. 11

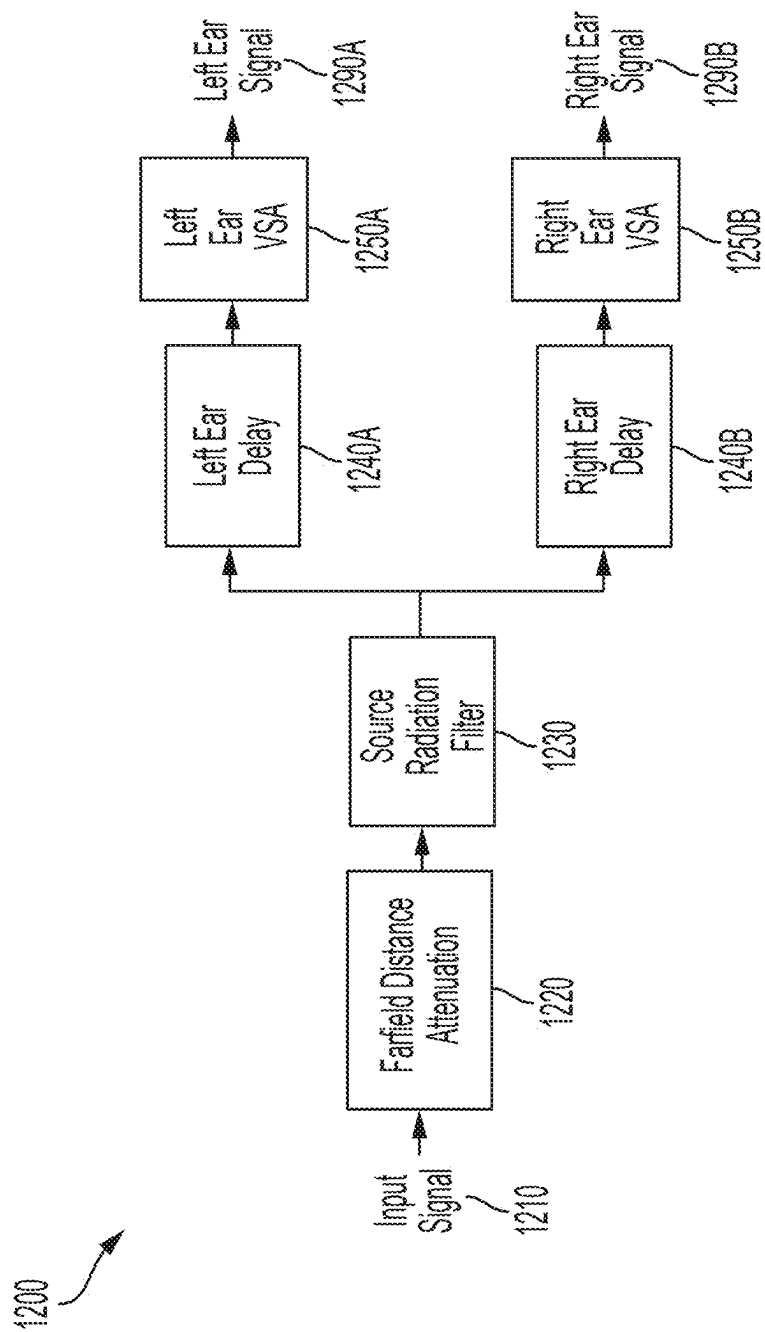


FIG. 12

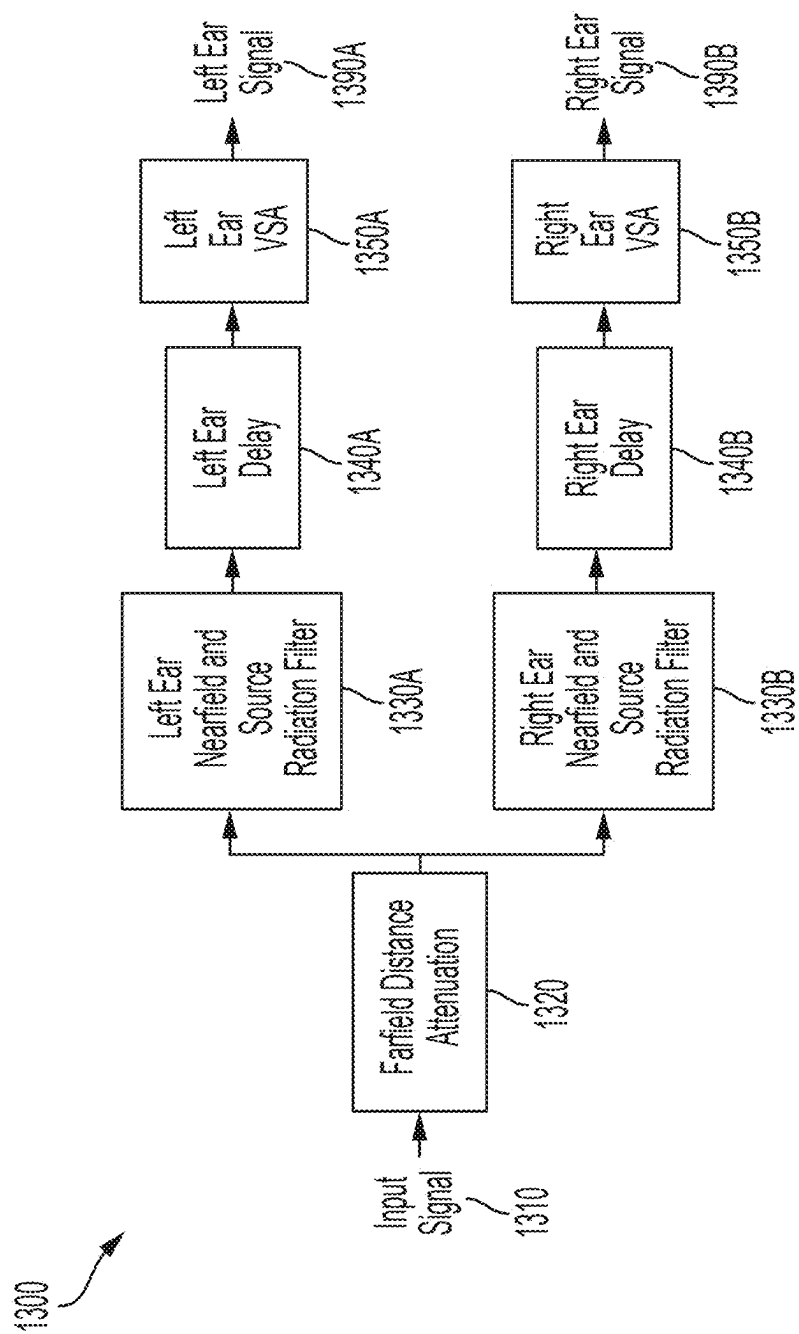


FIG. 13

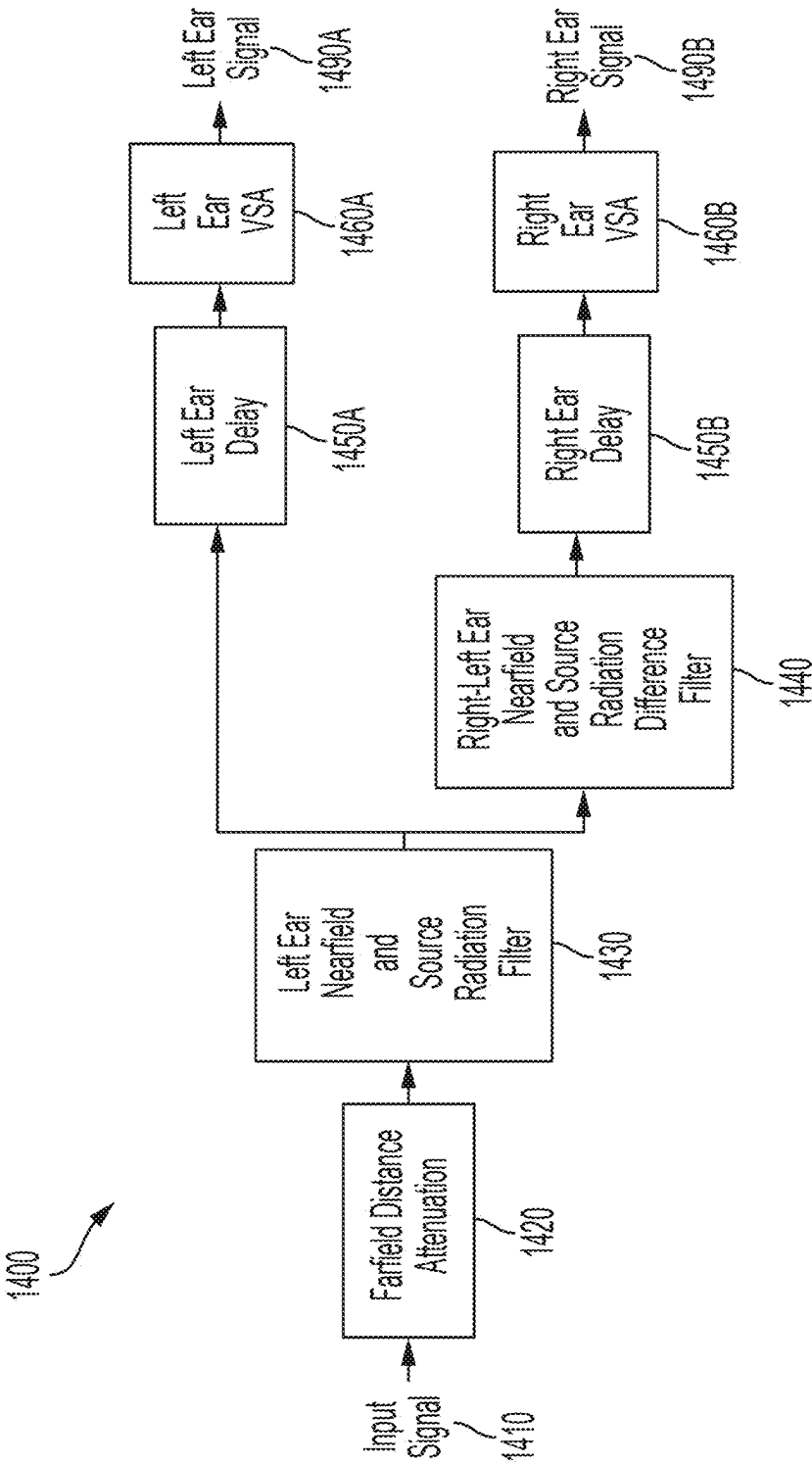


FIG. 14

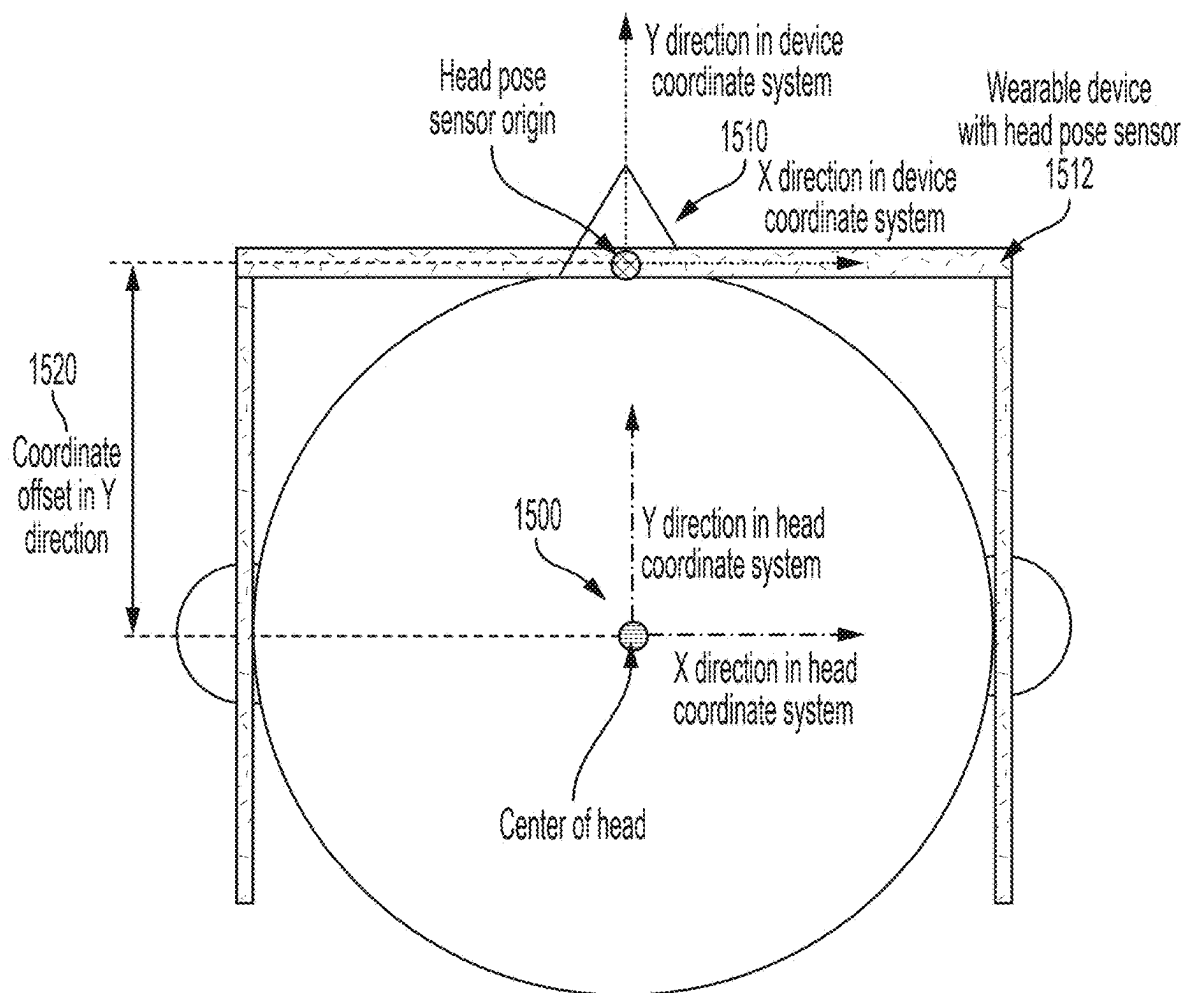


FIG. 15A

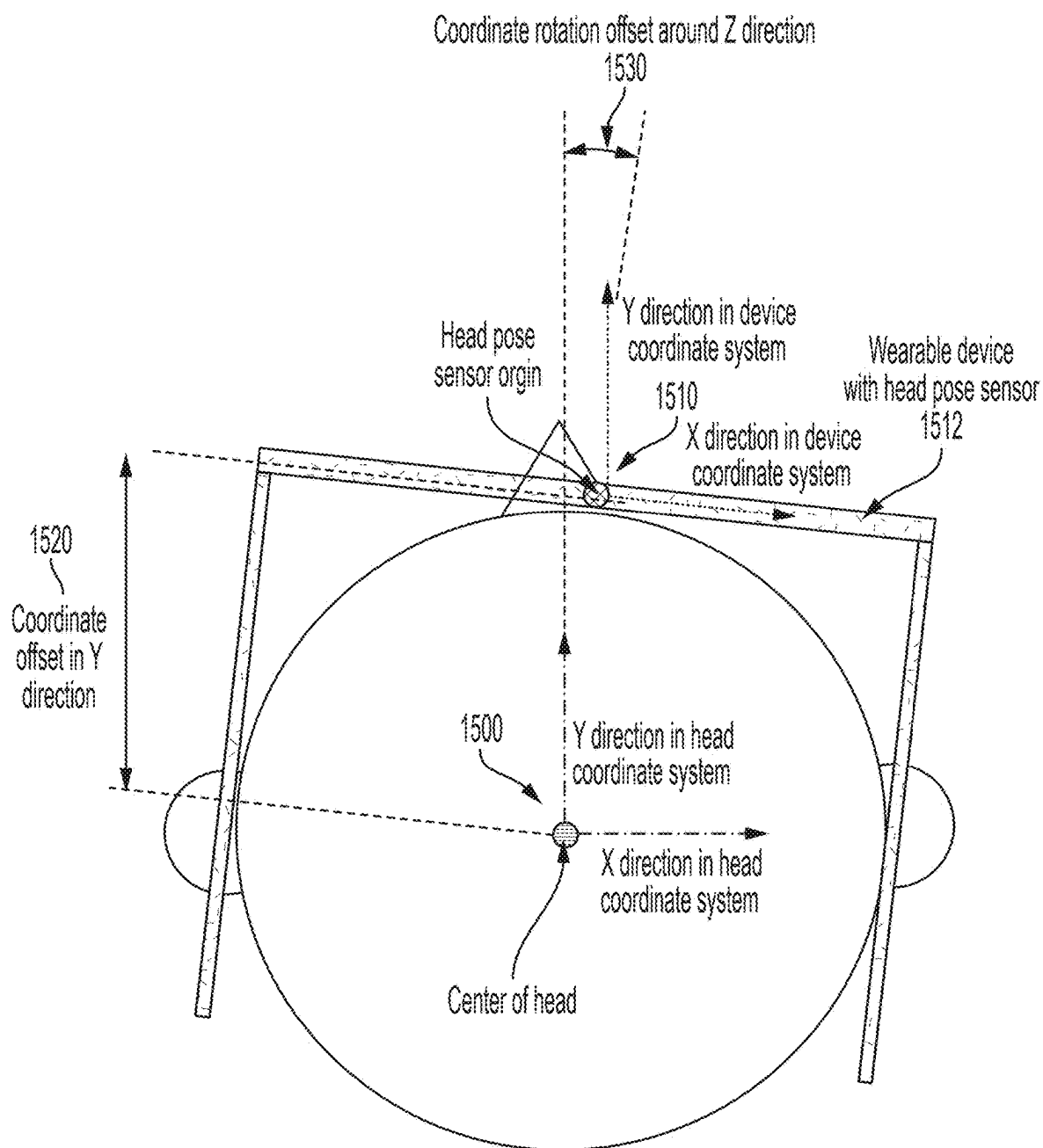


FIG. 15B

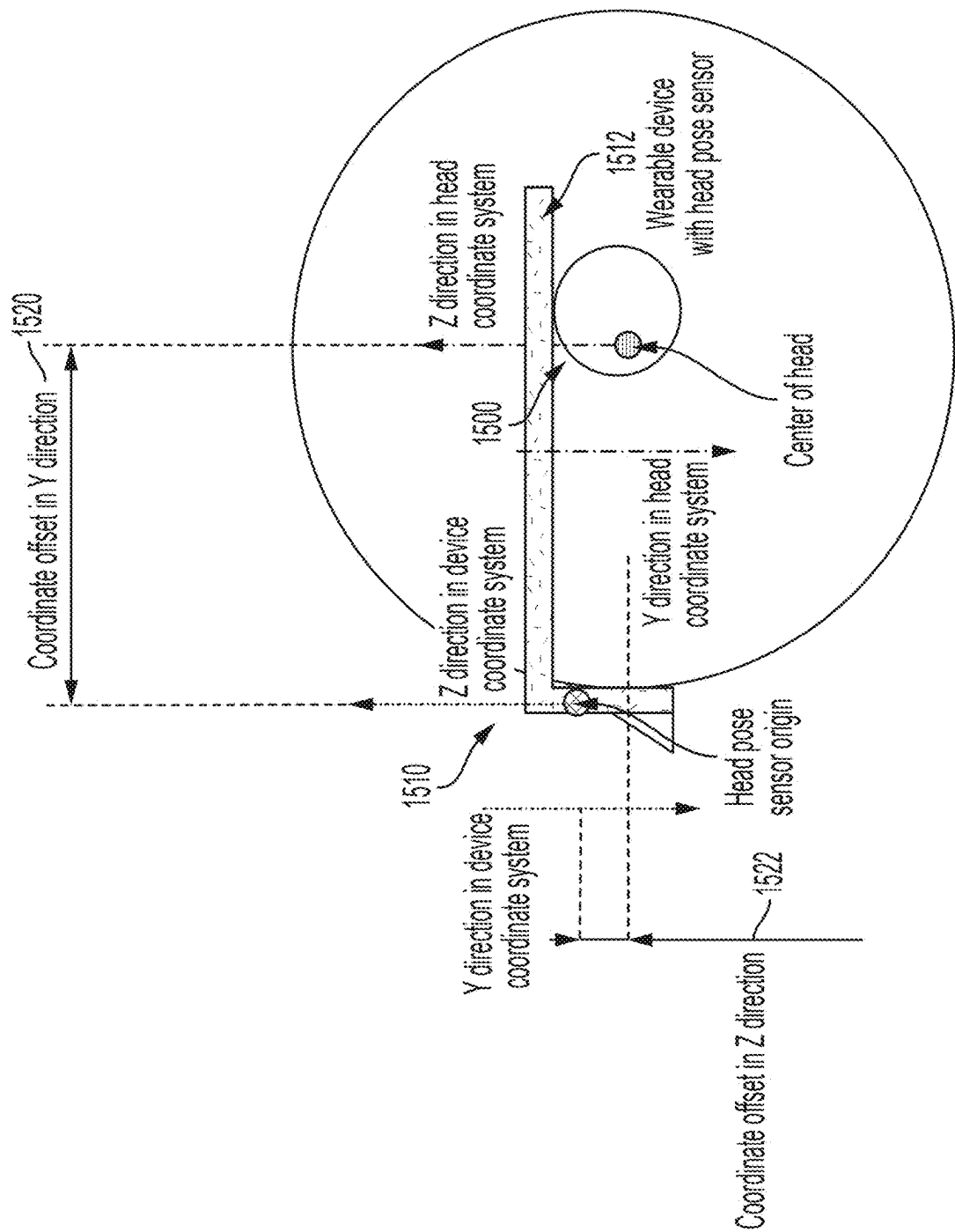
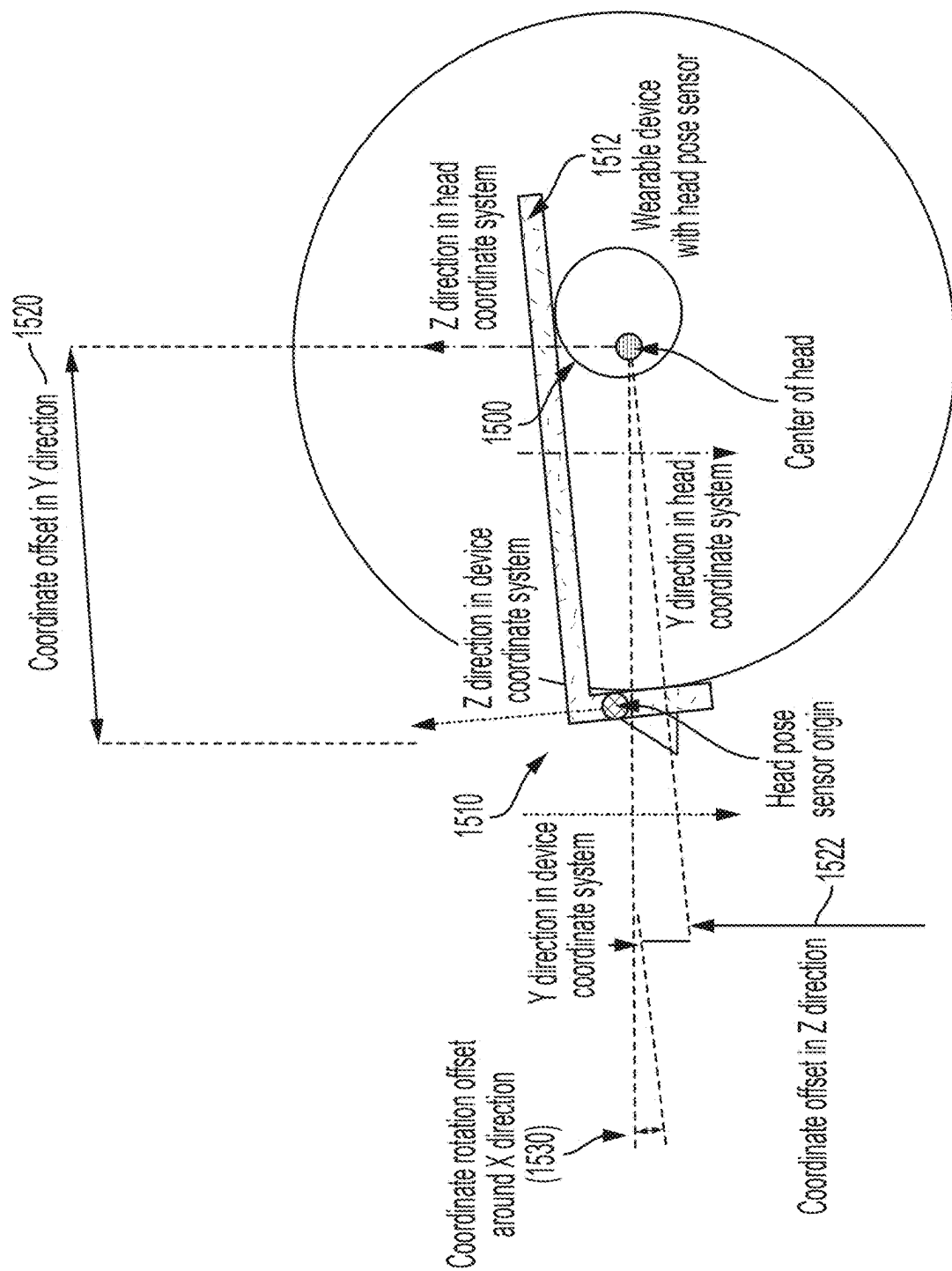


FIG. 15C



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E

NEAR-FIELD AUDIO RENDERING

REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/761,089, filed Jul. 1, 2024, which is a continuation of U.S. patent application Ser. No. 18/451,794, filed on Aug. 17, 2023, now U.S. Pat. No. 12,063,497, which is a continuation of U.S. patent application Ser. No. 18/061,367, filed on Dec. 2, 2022, now U.S. Pat. No. 11,778,411, which is a continuation of U.S. patent application Ser. No. 17/401,090, filed on Aug. 12, 2021, now U.S. Pat. No. 11,546,716, which is a continuation of U.S. patent application Ser. No. 16/593,943, filed on Oct. 4, 2019, now U.S. Pat. No. 11,122,383, which claims priority to U.S. Provisional Application No. 62/741,677, filed on Oct. 5, 2018, and to U.S. Provisional Application No. 62/812,734, filed on Mar. 1, 2019, the contents of which are incorporated by reference herein in their entirety.

FIELD

[0002] This disclosure relates generally to systems and methods for audio signal processing, and in particular to systems and methods for presenting audio signals in a mixed reality environment.

BACKGROUND

[0003] Augmented reality and mixed reality systems place unique demands on the presentation of binaural audio signals to a user. On one hand, presentation of audio signals in a realistic manner—for example, in a manner consistent with the user's expectations—is crucial for creating augmented or mixed reality environments that are immersive and believable. On the other hand, the computational expense of processing such audio signals can be prohibitive, particularly for mobile systems that may feature limited processing power and battery capacity.

[0004] One particular challenge is the simulation of near-field audio effects. Near-field effects are important for recreating impression of a sound source coming very close to a user's head. Near-field effects can be computed using databases of head-related transfer functions (HRTFs). However, typical HRTF databases include HRTFs measured at a single distance in a far-field from the user's head (e.g., more than 1 meter from the user's head), and may lack HRTFs at distances suitable for near-field effects. And even if the HRTF databases included measured or simulated HRTFs for different distances from the user's head (e.g., less than 1 meter from the user's head), it may be computationally expensive to directly use a high number of HRTFs for real-time audio rendering applications. Accordingly, systems and methods are desired for modeling near-field audio effects using far-field HRTFs in a computationally efficient manner.

BRIEF SUMMARY

[0005] Examples of the disclosure describe systems and methods for presenting an audio signal to a user of a wearable head device. According to an example method, a source location corresponding to the audio signal is identified. An acoustic axis corresponding to the audio signal is determined. For each of a respective left and right ear of the user, an angle between the acoustic axis and the respective ear is determined. For each of the respective left and right

ear of the user, a virtual speaker position, of a virtual speaker array, is determined, the virtual speaker position collinear with the source location and with a position of the respective ear. The virtual speaker array comprises a plurality of virtual speaker positions, each virtual speaker position of the plurality located on the surface of a sphere concentric with the user's head, the sphere having a first radius. For each of the respective left and right ear of the user, a head-related transfer function (HRTF) corresponding to the virtual speaker position and to the respective ear is determined; a source radiation filter is determined based on the determined angle; the audio signal is processed to generate an output audio signal for the respective ear; and the output audio signal is presented to the respective ear of the user via one or more speakers associated with the wearable head device. Processing the audio signal comprises applying the HRTF and the source radiation filter to the audio signal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 illustrates an example wearable system, according to some embodiments of the disclosure.

[0007] FIG. 2 illustrates an example handheld controller that can be used in conjunction with an example wearable system, according to some embodiments of the disclosure.

[0008] FIG. 3 illustrates an example auxiliary unit that can be used in conjunction with an example wearable system, according to some embodiments of the disclosure.

[0009] FIG. 4 illustrates an example functional block diagram for an example wearable system, according to some embodiments of the disclosure.

[0010] FIG. 5 illustrates a binaural rendering system, according to some embodiments of the disclosure.

[0011] FIGS. 6A-6C illustrate example geometry of modeling audio effects from a virtual sound source, according to some embodiments of the disclosure.

[0012] FIG. 7 illustrates an example of computing a distance traveled by sound emitted by a point sound source, according to some embodiments of the disclosure.

[0013] FIGS. 8A-8C illustrate examples of a sound source relative to an ear of a listener, according to some embodiments of the disclosure.

[0014] FIGS. 9A-9B illustrate example Head-Related Transfer Function (HRTF) magnitude responses, according to some embodiments of the disclosure.

[0015] FIG. 10 illustrates a source radiation angle of a user relative to an acoustical axis of a sound source, according to some embodiments of the disclosure.

[0016] FIG. 11 illustrates an example of a sound source panned inside a user's head, according to some embodiments of the disclosure.

[0017] FIG. 12 illustrates an example signal flow that may be implemented to render a sound source in a far-field, according to some embodiments of the disclosure.

[0018] FIG. 13 illustrates an example signal flow that may be implemented to render a sound source in a near-field, according to some embodiments of the disclosure.

[0019] FIG. 14 illustrates an example signal flow that may be implemented to render a sound source in a near-field, according to some embodiments of the disclosure.

[0020] FIGS. 15A-15D illustrate examples of a head coordinate system corresponding to a user and a device coordinate system corresponding to a device, according to some embodiments of the disclosure.

DETAILED DESCRIPTION

[0021] In the following description of examples, reference is made to the accompanying drawings which form a part hereof, and in which it is shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the disclosed examples.

Example Wearable System

[0022] FIG. 1 illustrates an example wearable head device **100** configured to be worn on the head of a user. Wearable head device **100** may be part of a broader wearable system that includes one or more components, such as a head device (e.g., wearable head device **100**), a handheld controller (e.g., handheld controller **200** described below), and/or an auxiliary unit (e.g., auxiliary unit **300** described below). In some examples, wearable head device **100** can be used for virtual reality, augmented reality, or mixed reality systems or applications. Wearable head device **100** can include one or more displays, such as displays **110A** and **110B** (which may include left and right transmissive displays, and associated components for coupling light from the displays to the user's eyes, such as orthogonal pupil expansion (OPE) grating sets **112A/112B** and exit pupil expansion (EPE) grating sets **114A/114B**); left and right acoustic structures, such as speakers **120A** and **120B** (which may be mounted on temple arms **122A** and **122B**, and positioned adjacent to the user's left and right ears, respectively); one or more sensors such as infrared sensors, accelerometers, GPS units, inertial measurement units (IMUs, e.g. IMU **126**), acoustic sensors (e.g., microphones **150**); orthogonal coil electromagnetic receivers (e.g., receiver **127** shown mounted to the left temple arm **122A**); left and right cameras (e.g., depth (time-of-flight) cameras **130A** and **130B**) oriented away from the user; and left and right eye cameras oriented toward the user (e.g., for detecting the user's eye movements) (e.g., eye cameras **128A** and **128B**). However, wearable head device **100** can incorporate any suitable display technology, and any suitable number, type, or combination of sensors or other components without departing from the scope of the disclosure. In some examples, wearable head device **100** may incorporate one or more microphones **150** configured to detect audio signals generated by the user's voice; such microphones may be positioned adjacent to the user's mouth. In some examples, wearable head device **100** may incorporate networking features (e.g., Wi-Fi capability) to communicate with other devices and systems, including other wearable systems. Wearable head device **100** may further include components such as a battery, a processor, a memory, a storage unit, or various input devices (e.g., buttons, touchpads); or may be coupled to a handheld controller (e.g., handheld controller **200**) or an auxiliary unit (e.g., auxiliary unit **300**) that comprises one or more such components. In some examples, sensors may be configured to output a set of coordinates of the head-mounted unit relative to the user's environment, and may provide input to a processor performing a Simultaneous Localization and Mapping (SLAM) procedure and/or a visual odometry algorithm. In some examples, wearable head device **100** may be coupled to a handheld controller **200**, and/or an auxiliary unit **300**, as described further below.

[0023] FIG. 2 illustrates an example mobile handheld controller component **200** of an example wearable system.

In some examples, handheld controller **200** may be in wired or wireless communication with wearable head device **100** and/or auxiliary unit **300** described below. In some examples, handheld controller **200** includes a handle portion **220** to be held by a user, and one or more buttons **240** disposed along a top surface **210**. In some examples, handheld controller **200** may be configured for use as an optical tracking target; for example, a sensor (e.g., a camera or other optical sensor) of wearable head device **100** can be configured to detect a position and/or orientation of handheld controller **200**—which may, by extension, indicate a position and/or orientation of the hand of a user holding handheld controller **200**. In some examples, handheld controller **200** may include a processor, a memory, a storage unit, a display, or one or more input devices, such as described above. In some examples, handheld controller **200** includes one or more sensors (e.g., any of the sensors or tracking components described above with respect to wearable head device **100**). In some examples, sensors can detect a position or orientation of handheld controller **200** relative to wearable head device **100** or to another component of a wearable system. In some examples, sensors may be positioned in handle portion **220** of handheld controller **200**, and/or may be mechanically coupled to the handheld controller. Handheld controller **200** can be configured to provide one or more output signals, corresponding, for example, to a pressed state of the buttons **240**; or a position, orientation, and/or motion of the handheld controller **200** (e.g., via an IMU). Such output signals may be used as input to a processor of wearable head device **100**, to auxiliary unit **300**, or to another component of a wearable system. In some examples, handheld controller **200** can include one or more microphones to detect sounds (e.g., a user's speech, environmental sounds), and in some cases provide a signal corresponding to the detected sound to a processor (e.g., a processor of wearable head device **100**).

[0024] FIG. 3 illustrates an example auxiliary unit **300** of an example wearable system. In some examples, auxiliary unit **300** may be in wired or wireless communication with wearable head device **100** and/or handheld controller **200**. The auxiliary unit **300** can include a battery to provide energy to operate one or more components of a wearable system, such as wearable head device **100** and/or handheld controller **200** (including displays, sensors, acoustic structures, processors, microphones, and/or other components of wearable head device **100** or handheld controller **200**). In some examples, auxiliary unit **300** may include a processor, a memory, a storage unit, a display, one or more input devices, and/or one or more sensors, such as described above. In some examples, auxiliary unit **300** includes a clip **310** for attaching the auxiliary unit to a user (e.g., a belt worn by the user). An advantage of using auxiliary unit **300** to house one or more components of a wearable system is that doing so may allow large or heavy components to be carried on a user's waist, chest, or back—which are relatively well suited to support large and heavy objects—rather than mounted to the user's head (e.g., if housed in wearable head device **100**) or carried by the user's hand (e.g., if housed in handheld controller **200**). This may be particularly advantageous for relatively heavy or bulky components, such as batteries.

[0025] FIG. 4 shows an example functional block diagram that may correspond to an example wearable system **400**, such as may include example wearable head device **100**,

handheld controller **200**, and auxiliary unit **300** described above. In some examples, the wearable system **400** could be used for virtual reality, augmented reality, or mixed reality applications. As shown in FIG. 4, wearable system **400** can include example handheld controller **400B**, referred to here as a “totem” (and which may correspond to handheld controller **200** described above); the handheld controller **400B** can include a totem-to-headgear six degree of freedom (6DOF) totem subsystem **404A**. Wearable system **400** can also include example headgear device **400A** (which may correspond to wearable head device **100** described above); the headgear device **400A** includes a totem-to-headgear 6DOF headgear subsystem **404B**. In the example, the 6DOF totem subsystem **404A** and the 6DOF headgear subsystem **404B** cooperate to determine six coordinates (e.g., offsets in three translation directions and rotation along three axes) of the handheld controller **400B** relative to the headgear device **400A**. The six degrees of freedom may be expressed relative to a coordinate system of the headgear device **400A**. The three translation offsets may be expressed as X, Y, and Z offsets in such a coordinate system, as a translation matrix, or as some other representation. The rotation degrees of freedom may be expressed as sequence of yaw, pitch and roll rotations; as vectors; as a rotation matrix; as a quaternion; or as some other representation. In some examples, one or more depth cameras **444** (and/or one or more non-depth cameras) included in the headgear device **400A**; and/or one or more optical targets (e.g., buttons **240** of handheld controller **200** as described above, or dedicated optical targets included in the handheld controller) can be used for 6DOF tracking. In some examples, the handheld controller **400B** can include a camera, as described above; and the headgear device **400A** can include an optical target for optical tracking in conjunction with the camera. In some examples, the headgear device **400A** and the handheld controller **400B** each include a set of three orthogonally oriented solenoids which are used to wirelessly send and receive three distinguishable signals. By measuring the relative magnitude of the three distinguishable signals received in each of the coils used for receiving, the 6DOF of the handheld controller **400B** relative to the headgear device **400A** may be determined. In some examples, 6DOF totem subsystem **404A** can include an Inertial Measurement Unit (IMU) that is useful to provide improved accuracy and/or more timely information on rapid movements of the handheld controller **400B**.

[0026] In some examples involving augmented reality or mixed reality applications, it may be desirable to transform coordinates from a local coordinate space (e.g., a coordinate space fixed relative to headgear device **400A**) to an inertial coordinate space, or to an environmental coordinate space. For instance, such transformations may be necessary for a display of headgear device **400A** to present a virtual object at an expected position and orientation relative to the real environment (e.g., a virtual person sitting in a real chair, facing forward, regardless of the position and orientation of headgear device **400A**), rather than at a fixed position and orientation on the display (e.g., at the same position in the display of headgear device **400A**). This can maintain an illusion that the virtual object exists in the real environment (and does not, for example, appear positioned unnaturally in the real environment as the headgear device **400A** shifts and rotates). In some examples, a compensatory transformation between coordinate spaces can be determined by processing

imagery from the depth cameras **444** (e.g., using a Simultaneous Localization and Mapping (SLAM) and/or visual odometry procedure) in order to determine the transformation of the headgear device **400A** relative to an inertial or environmental coordinate system. In the example shown in FIG. 4, the depth cameras **444** can be coupled to a SLAM/visual odometry block **406** and can provide imagery to block **406**. The SLAM/visual odometry block **406** implementation can include a processor configured to process this imagery and determine a position and orientation of the user’s head, which can then be used to identify a transformation between a head coordinate space and a real coordinate space. Similarly, in some examples, an additional source of information on the user’s head pose and location is obtained from an IMU **409** of headgear device **400A**. Information from the IMU **409** can be integrated with information from the SLAM/visual odometry block **406** to provide improved accuracy and/or more timely information on rapid adjustments of the user’s head pose and position.

[0027] In some examples, the depth cameras **444** can supply 3D imagery to a hand gesture tracker **411**, which may be implemented in a processor of headgear device **400A**. The hand gesture tracker **411** can identify a user’s hand gestures, for example by matching 3D imagery received from the depth cameras **444** to stored patterns representing hand gestures. Other suitable techniques of identifying a user’s hand gestures will be apparent.

[0028] In some examples, one or more processors **416** may be configured to receive data from headgear subsystem **404B**, the IMU **409**, the SLAM/visual odometry block **406**, depth cameras **444**, microphones **450**; and/or the hand gesture tracker **411**. The processor **416** can also send and receive control signals from the 6DOF totem system **404A**. The processor **416** may be coupled to the 6DOF totem system **404A** wirelessly, such as in examples where the handheld controller **400B** is untethered. Processor **416** may further communicate with additional components, such as an audio-visual content memory **418**, a Graphical Processing Unit (GPU) **420**, and/or a Digital Signal Processor (DSP) audio spatializer **422**. The DSP audio spatializer **422** may be coupled to a Head Related Transfer Function (HRTF) memory **425**. The GPU **420** can include a left channel output coupled to the left source of imagewise modulated light **424** and a right channel output coupled to the right source of imagewise modulated light **426**. GPU **420** can output stereoscopic image data to the sources of imagewise modulated light **424**, **426**. The DSP audio spatializer **422** can output audio to a left speaker **412** and/or a right speaker **414**. The DSP audio spatializer **422** can receive input from processor **419** indicating a direction vector from a user to a virtual sound source (which may be moved by the user, e.g., via the handheld controller **400B**). Based on the direction vector, the DSP audio spatializer **422** can determine a corresponding HRTF (e.g., by accessing a HRTF, or by interpolating multiple HRTFs). The DSP audio spatializer **422** can then apply the determined HRTF to an audio signal, such as an audio signal corresponding to a virtual sound generated by a virtual object. This can enhance the believability and realism of the virtual sound, by incorporating the relative position and orientation of the user relative to the virtual sound in the mixed reality environment—that is, by presenting a virtual sound that matches a user’s expectations of what that virtual sound would sound like if it were a real sound in a real environment.

[0029] In some examples, such as shown in FIG. 4, one or more of processor 416, GPU 420, DSP audio spatializer 422, HRTF memory 425, and audio/visual content memory 418 may be included in an auxiliary unit 400C (which may correspond to auxiliary unit 300 described above). The auxiliary unit 400C may include a battery 427 to power its components and/or to supply power to headgear device 400A and/or handheld controller 400B. Including such components in an auxiliary unit, which can be mounted to a user's waist, can limit the size and weight of headgear device 400A, which can in turn reduce fatigue of a user's head and neck.

[0030] While FIG. 4 presents elements corresponding to various components of an example wearable system 400, various other suitable arrangements of these components will become apparent to those skilled in the art. For example, elements presented in FIG. 4 as being associated with auxiliary unit 400C could instead be associated with headgear device 400A or handheld controller 400B. Furthermore, some wearable systems may forgo entirely a handheld controller 400B or auxiliary unit 400C. Such changes and modifications are to be understood as being included within the scope of the disclosed examples.

Audio Rendering

[0031] The systems and methods described below can be implemented in an augmented reality or mixed reality system, such as described above. For example, one or more processors (e.g., CPUs, DSPs) of an augmented reality system can be used to process audio signals or to implement steps of computer-implemented methods described below; sensors of the augmented reality system (e.g., cameras, acoustic sensors, IMUs, LIDAR, GPS) can be used to determine a position and/or orientation of a user of the system, or of elements in the user's environment; and speakers of the augmented reality system can be used to present audio signals to the user. In some embodiments, external audio playback devices (e.g. headphones, earbuds) could be used instead of the system's speakers for delivering the audio signal to the user's ears.

[0032] In augmented reality or mixed reality systems such as described above, one or more processors (e.g., DSP audio spatializer 422) can process one or more audio signals for presentation to a user of a wearable head device via one or more speakers (e.g., left and right speakers 412/414 described above). Processing of audio signals requires tradeoffs between the authenticity of a perceived audio signal—for example, the degree to which an audio signal presented to a user in a mixed reality environment matches the user's expectations of how an audio signal would sound in a real environment—and the computational overhead involved in processing the audio signal.

[0033] Modeling near-field audio effects can improve the authenticity of a user's audio experience, but can be computationally prohibitive. In some embodiments, an integrated solution may combine a computationally efficient rendering approach with one or more near-field effects for each ear. The one or more near-field effects for each ear may include, for example, parallax angles in simulation of sound incident for each ear, interaural time difference (ITDs) based on object position and anthropometric data, near-field level changes due to distance, and/or magnitude response changes due to proximity to the user's head and/or source radiation variation due to parallax angles. In some embodiments, the

integrated solution may be computationally efficient so as to not excessively increase computational cost.

[0034] In a far-field, as a sound source moves closer or farther from a user, changes at the user's ears may be the same for each ear and may be an attenuation of a signal for the sound source. In a near-field, as a sound source moves closer or farther from the user, changes at the user's ears may be different for each ear and may be more than just attenuations of the signal for the sound source. In some embodiments, the near-field and far-field boundaries may be where the conditions change.

[0035] In some embodiments, a virtual speaker array (VSA) may be a discrete set of positions on a sphere centered at a center of the user's head. For each position on the sphere, a pair (e.g., left-right pair) of HRTFs is provided. In some embodiments, a near-field may be a region inside the VSA and a far-field may be a region outside the VSA. At the VSA, either a near-field approach or a far-field approach may be used.

[0036] A distance from a center of the user's head to a VSA may be a distance at which the HRTFs were obtained. For example, the HRTF filters may be measured or synthesized from simulation. The measured/simulated distance from the VSA to the center of the user's head may be referred to as “measured distance” (MD). A distance from a virtual sound source to the center of the user's head may be referred to as “source distance” (SD).

[0037] FIG. 5 illustrates a binaural rendering system 500, according to some embodiments. In the example system of FIG. 5, a mono input audio signal 501 (which can represent a virtual sound source) is split by an interaural time delay (ITD) module 502 of an encoder 503 into a left signal 504 and a right signal 506. In some examples, the left signal 504 and the right signal 506 may differ by an ITD (e.g., in milliseconds) determined by the ITD module 502. In the example, the left signal 504 is input to a left ear VSA module 510 and the right signal 506 is input to a right ear VSA module 520.

[0038] In the example, the left ear VSA module 510 can pan the left signal 504 over a set of N channels respectively feeding a set of left-ear HRTF filters 550 ($L_1, \dots, L_N$) in a HRTF filter bank 540. The left-ear HRTF filters 550 may be substantially delay-free. Panning gains 512 ($g_{L1}, \dots, g_{LN}$) of the left ear VSA module may be functions of a left incident angle (ang_L). The left incident angle may be indicative of a direction of incidence of sound relative to a frontal direction from the center of the user's head. Though shown from a top-down perspective with respect to the user's head in the figure, the left incident angle can comprise an angle in three dimensions; that is, the left incident angle can include an azimuth and/or an elevation angle.

[0039] Similarly, in the example, the right ear VSA module 520 can pan the right signal 506 over a set of M channels respectively feeding a set of right-ear HRTF filters 560 ($R_1, \dots, R_M$) in the HRTF filter bank 540. The right-ear HRTF filters 550 may be substantially delay-free. (Although only one HRTF filter bank is shown in the figure, multiple HRTF filter banks, including those stored across distributed systems, are contemplated.) Panning gains 522 ($g_{R1}, \dots, g_{RM}$) of the right ear VSA module may be functions of a right incident angle (ang_R). The right incident angle may be indicative of a direction of incidence of sound relative to the frontal direction from the center of the user's head. As above, the right incident angle can comprise an angle in

three dimensions; that is, the right incident angle can include an azimuth and/or an elevation angle.

[0040] In some embodiments, such as shown, the left ear VSA module 510 may pan the left signal 504 over N channels and the right ear VSA module may pan the right signal over M channels. In some embodiments, N and M may be equal. In some embodiments, N and M may be different. In these embodiments, the left ear VSA module may feed into a set of left-ear HRTF filters ($L_1, \dots, L_N$) and the right ear VSA module may feed into a set of right-ear HRTF filters ($R_1, \dots, R_M$), as described above. Further, in these embodiments, panning gains ($g_{L1}, \dots, g_{LN}$) of the left ear VSA module may be functions of a left ear incident angle (ang_L) and panning gains ($g_{R1}, \dots, g_{RM}$) of the right ear VSA module may be functions of a right ear incident angle (ang_R), as described above.

[0041] The example system illustrates a single encoder 503 and corresponding input signal 501. The input signal may correspond to a virtual sound source. In some embodiments, the system may include additional encoders and corresponding input signals. In these embodiments, the input signals may correspond to virtual sound sources. That is, each input signal may correspond to a virtual sound source.

[0042] In some embodiments, when simultaneously rendering several virtual sound sources, the system may include an encoder per virtual sound source. In these embodiments, a mix module (e.g., 530 in FIG. 5) receives outputs from each of the encoders, mixes the received signals, and outputs mixed signals to the left and right HRTF filters of the HRTF filter bank.

[0043] FIG. 6A illustrates a geometry for modeling audio effects from a virtual sound source, according to some embodiments. A distance 630 of the virtual sound source 610 to a center 620 of a user's head (e.g., "source distance" (SD)) is equal to a distance 640 from a VSA 650 to the center of the user's head (e.g., "measured distance" (MD)). As illustrated in FIG. 6A, a left incident angle 652 (ang_L) and a right incident angle 654 (ang_R) are equal. In some embodiments, an angle from the center 620 of the user's head to the virtual sound source 610 may be used directly for computing panning gains (e.g., $g_{L1}, \dots, g_{LN}, g_{R1}, \dots, g_{RM}$). In the example shown, the virtual sound source position 610 is used as the position (612/614) for computing left ear panning and right ear panning.

[0044] FIG. 6B illustrates a geometry for modeling near-field audio effects from a virtual sound source, according to some embodiments. As shown, a distance 630 from the virtual sound source 610 to a reference point (e.g., "source distance" (SD)) is less than a distance 640 from a VSA 650 to the center 620 of the user's head (e.g., "measured distance" (MD)). In some embodiments, the reference point may be a center of a user's head (620). In some embodiments, the reference point may be a mid-point between two cars of the user. As illustrated in FIG. 6B, a left incident angle 652 (ang_L) is greater than a right incident angle 654 (ang_R). Angles relative to each ear (e.g., the left incident angle 652 (ang_L) and the right incident angle 654 (ang_R)) are different than at the MD 640.

[0045] In some embodiments, the left incident angle 652 (ang_L) used for computing a left ear signal panning may be derived by computing an intersection of a line going from the user's left ear through a location of the virtual sound source 610, and a sphere containing the VSA 650. A panning

angle combination (azimuth and elevation) may be computed for 3D environments as a spherical coordinate angle from the center 620 of the user's head to the intersection point.

[0046] Similarly, in some embodiments, the right incident angle 654 (ang_R) used for computing a right ear signal panning may be derived by computing an intersection of a line going from the user's right ear through the location of the virtual sound source 610, and the sphere containing the VSA 650. A panning angle combination (azimuth and elevation) may be computed for 3D environments as a spherical coordinate angle from the center 620 of the user's head to the intersection point.

[0047] In some embodiments, an intersection between a line and a sphere may be computed, for example, by combining an equation representing the line and an equation representing the sphere.

[0048] FIG. 6C illustrates a geometry for modeling far-field audio effects from a virtual sound source, according to some embodiments. A distance 630 of the virtual sound source 610 to a center 620 of a user's head (e.g., "source distance" (SD)) is greater than a distance 640 from a VSA 650 to the center 620 of the user's head (e.g., "measured distance" (MD)). As illustrated in FIG. 6C, a left incident angle 612 (ang_L) is less than a right incident angle 614 (ang_R). Angles relative to each ear (e.g., the left incident angle (ang_L) and the right incident angle (ang_R)) are different than at the MD.

[0049] In some embodiments, the left incident angle 612 (ang_L) used for computing a left ear signal panning may be derived by computing an intersection of a line going from the user's left ear through a location of the virtual sound source 610, and a sphere containing the VSA 650. A panning angle combination (azimuth and elevation) may be computed for 3D environments as a spherical coordinate angle from the center 620 of the user's head to the intersection point.

[0050] Similarly, in some embodiments, the right incident angle 614 (ang_R) used for computing a right ear signal panning may be derived by computing an intersection of a line going from the user's right ear through the location of the virtual sound source 610, and the sphere containing the VSA 650. A panning angle combination (azimuth and elevation) may be computed for 3D environments as a spherical coordinate angle from the center 620 of the user's head to the intersection point.

[0051] In some embodiments, an intersection between a line and a sphere may be computed, for example, by combining an equation representing the line and an equation representing the sphere.

[0052] In some embodiments, rendering schemes may not differentiate the left incident angle 612 and the right incident angle 614, and instead assume the left incident angle 612 and the right incident angle 614 are equal. However, assuming the left incident angle 612 and the right incident angle 614 are equal may not be applicable or acceptable when reproducing near-field effects as described with respect to FIG. 6B and/or far-field effects as described with respect to FIG. 6C.

[0053] FIG. 7 illustrates a geometric model for computing a distance traveled by sound emitted by a (point) sound source 710 to an ear 712 of the user, according to some embodiments. In the geometric model illustrated in FIG. 7, a user's head is assumed to be spherical. A same model is

applied to each ear (e.g., a left ear and a right ear). A delay to each ear may be computed by dividing a distance travelled by sound emitted by the (point) sound source **710** to the ear **712** of the user (e.g., distance A+B in FIG. 7) by the speed of sound in the user's environment (e.g., air). An interaural time difference (ITD) may be a difference in delay between the user's two ears. In some embodiments, the ITD may be applied to only a contralateral ear with respect to the user's head and a location of the sound source **710**. In some embodiments, the geometric model illustrated in FIG. 7 may be used for any SD (e.g., near-field or far-field) and may not take into account positions of the ears on the user's head and/or head size of the user's head.

[0054] In some embodiments, the geometric model illustrated in FIG. 7 may be used to compute attenuation due to a distance from a sound source **710** to each ear. In some embodiments, the attenuation may be computed using a ratio of distances. A difference in level for near-field sources may be computed by evaluating a ratio of a source-to-ear distance for a desired source position, and a source-to-ear distance for a source corresponding to the MD and angles computed for panning (e.g., as illustrated in FIGS. 6A-6C). In some embodiments, a minimum distance from the ears may be used, for example, to avoid dividing by very small numbers which may be computationally expensive and/or result in numerical overflow. In these embodiments, smaller distances may be clamped.

[0055] In some embodiments, distances may be clamped. Clamping may include, for example, limiting distance values below a threshold value to another value. In some embodiments, clamping may include using the limited distance values (referred to as clamped distance values), instead of the actual distance values, for computations. Hard clamping may include limiting distance values below a threshold value to the threshold value. For example, if a threshold value is 5 millimeters, then distance values less than the threshold value will be set to the threshold value, and the threshold value, instead of the actual distance value which is less than the threshold value, may be used for computations. Soft clamping may include limiting distance values such that as the distance values approach or go below a threshold value, they asymptotically approach the threshold value. In some embodiments, instead of, or in addition to, clamping, distance values may be increased by a predetermined amount such that the distance values are never less than the predetermined amount.

[0056] In some embodiments, a first minimum distance from the ears of the listener may be used for computing gains and a second minimum distance from the ears of the listener may be used for computing other sound source position parameters such as, for example, angles used for computing HRTF filters, interaural time differences, and the like. In some embodiments, the first minimum distance and the second minimum distance may be different.

[0057] In some embodiments, the minimum distance used for computing gains may be a function of one or more properties of the sound source. In some embodiments, the minimum distance used for computing gains may be a function of a level (e.g., RMS value of a signal over a number of frames) of the sound source, a size of the sound source, or radiation properties of the sound source, and the like.

[0058] FIGS. 8A-8C illustrate examples of a sound source relative to a right ear of the listener, according to some

embodiments. FIG. 8A illustrates the case where the sound source **810** is at a distance **812** from the right ear **820** of the listener that is greater than the first minimum distance **822** and the second minimum distance **824**. In this embodiment, the distance **812** between the simulated sound source and the right ear **820** of the listener is used for computing gains and other sound source position parameters, and is not clamped.

[0059] FIG. 8B shows the case where the simulated sound source **810** is at a distance **812** from the right ear **820** of the listener that is less than the first minimum distance **822** and greater than the second minimum distance **824**. In this embodiment, the distance **812** is clamped for gain computation, but not for computing other parameters such as, for example, azimuth and elevation angles or interaural time differences. In other words, the first minimum distance **822** is used for computing gains, and the distance **812** between the simulated sound source **810** and the right ear **820** of the listener is used for computing other sound source position parameters.

[0060] FIG. 8C shows the case where the simulated sound source **810** is closer to the ear than both the first minimum distance **822** and the second minimum distance **824**. In this embodiment, the distance **812** is clamped for gain computation and for computing other sound source position parameters. In other words, the first minimum distance **822** is used for computing gains, and the second minimum distance **824** is used for computing other sound source position parameters.

[0061] In some embodiments, gains computed from distance may be limited directly in lieu of limiting minimum distance used to compute gains. In other words, the gain may be computed based on distance as a first step, and in a second step the gain may be clamped to not exceed a predetermined threshold value.

[0062] In some embodiments, as a sound source gets closer to the head of the listener, a magnitude response of the sound source may change. For example, as a sound source gets closer to the head of the listener, low frequencies at an ipsilateral ear may be amplified and/or high frequencies at a contralateral ear may be attenuated. Changes in the magnitude response may lead to changes in interaural level differences (ILDs).

[0063] FIGS. 9A and 9B illustrate HRTF magnitude responses **900A** and **900B**, respectively, at an ear for a (point) sound source in a horizontal plane, according to some embodiments. The HRTF magnitude responses may be computed using a spherical head model as a function of azimuth angles. FIG. 9A illustrates a magnitude response **900A** for a (point) sound source in a far-field (e.g., one meter from the center of the user's head). FIG. 9B illustrates a magnitude response **900B** for a (point) sound source in a near-field (e.g., 0.25 meters from the center of the user's head). As illustrated in FIGS. 9A and 9B, a change in ILD may be most noticeable at low frequencies. In the far-field, the magnitude response for low frequency content may be constant (e.g., independent of angle of source azimuth). In the near-field, the magnitude response of low frequency content may be amplified for sound sources on a same side of the user's head/ear, which may lead to a higher ILD at low frequencies. In the near-field, the magnitude response of the high frequency content may be attenuated for sound sources on an opposite side of the user's head.

[0064] In some embodiments, changes in magnitude response may be taken into account by, for example, con-

sidering HRTF filters used in binaural rendering. In the case of a VSA, the HRTF filters may be approximated as HRTFs corresponding to a position used for computing right ear panning and a position used for computing left ear panning (e.g., as illustrated in FIG. 6B and FIG. 6C). In some embodiments, the HRTF filters may be computed using direct MD HRTFs. In some embodiments, the HRTF filters may be computed using panned spherical head model HRTFs. In some embodiments, compensation filters may be computed independent of a parallax HRTF angle.

[0065] In some embodiments, parallax HRTF angles may be computed and then used to compute more accurate compensation filters. For example, referring to FIG. 6B, a position used for computing left ear panning may be compared to a virtual sound source position for computing composition filters for the left ear, and a position used for computing right ear panning may be compared to a virtual sound source position for computing composition filters for the right ear.

[0066] In some embodiments, once attenuations due to distance have been taken into account, magnitude differences may be captured with additional signal processing. In some embodiments, the additional signal processing may consist of a gain, a low shelving filter, and a high shelving filter to be applied to each ear signal.

[0067] In some embodiments, a broadband gain may be computed for angles up to 120 degrees, for example, according to equation 1:

$$\text{gain_db} = 2.5 * \sin(\text{angleMD_deg} * 3/2) \quad (\text{Equation 1})$$

[0068] where angleMD_deg may be an angle of a corresponding HRTF at a MD, for example, relative to a position of an ear of the user. In some embodiments, angles other than 120 degrees may be used. In these embodiments, Equation 1 may be modified per the angle used.

[0069] In some embodiments, a broadband gain may be computed for angles greater than 120 degrees, for example, according to equation 2:

$$\text{gain_db} = 2.5 * \sin(180 + 3 * (\text{angleMD_deg} - 120)) \quad (\text{Equation 2})$$

[0070] In some embodiments, angles other than 120 degrees may be used. In these embodiments, Equation 2 may be modified per the angle used.

[0071] In some embodiments, a low shelving filter gain may be computed, for example, according to equation 3:

$$\text{lowshelfgain_db} = 2.5 * (e^{-\text{angleMD_deg}/65} - e^{-180/65}) \quad (\text{Equation 3})$$

[0072] In some embodiments, other angles may be used. In these embodiments, Equation 3 may be modified per the angle used.

[0073] In some embodiment, a high shelving filter gain may be computed for angles larger than 110 degrees, for example, according to equation 4:

$$\text{highshelfgain_db} = \quad (\text{Equation 4})$$

$$3.3 * (\cos((\text{angle_deg} * 180/\pi - 110) * 3) - 1)$$

[0074] where angle_deg may be an angle of the source, relative to the position of the ear of the user. In some embodiments, angles other than 110 degrees may be used. In these embodiments, Equation 4 may be modified per the angle used.

[0075] The aforementioned effects (e.g., gain, low shelving filter, and high shelving filter) may be attenuated as a function of distance. In some embodiments, a distance attenuation factor may be computed, for example, according to equation 5:

$$\text{distanceAttenuation} = \quad (\text{Equation 5})$$

$$(HR/(HR - MD)) * (1 - MD/\text{sourceDistance_clamped})$$

[0076] where HR is the head radius, MD is the measured distance, and sourceDistance_clamped is the source distance clamped to be at least as big as the head radius.

[0077] FIG. 10 illustrates an off-axis angle (or source radiation angle) of a user relative to an acoustical axis 1015 of a sound source 1010, according to some embodiments. In some embodiments, the source radiation angle may be used to evaluate a magnitude response of a direct path, for example, based on source radiation properties. In some embodiments, an off-axis angle may be different for each ear as the source moves closer to the user's head. In the figure, source radiation angle 1020 corresponds to the left ear; source radiation angle 1030 corresponds to the center of the head; and source radiation angle 1040 corresponds to the right ear. Different off-axis angles for each ear may lead to separate direct path processing for each ear.

[0078] FIG. 11 illustrates a sound source 1110 panned inside a user's head, according to some embodiments. In order to create an in-head effect, the sound source 1110 may be processed as a crossfade between a binaural render and a stereo render. In some embodiments, the binaural render may be created for a source 1112 located on or outside the user's head. In some embodiments, the location of the sound source 1112 may be defined as the intersection of a line going from the center 1120 of the user's head through the simulated sound position 1110, and the surface 1130 of the user's head. In some embodiment, the stereo render may be created using amplitude and/or time based panning techniques. In some embodiments, a time based panning technique may be used to time align a stereo signal and a binaural signal at each ear, for example, by applying an ITD to a contralateral ear. In some embodiments, the ITD and an ILD may be scaled down to zero as the sound source approaches the center 1120 of the user's head (i.e., as source distance 1150 approaches zero). In some embodiments, the crossfade between binaural and stereo may be computed, for example, based on the SD, and may be normalized by an approximate radius 1140 of the user's head.

[0079] In some embodiments, a filter (e.g., an EQ filter) may be applied for a sound source placed at the center of the user's head. The EQ filter may be used to reduce abrupt

timbre changes as the sound source moves through the user's head. In some embodiment, the EQ filter may be scaled to match a magnitude response at the surface of the user's head as the simulated sound source moves from the center of the user's head to the surface of the user's head, and thus further reduce a risk of abrupt magnitude response changes when the sound source moves in and out of the user's head. In some embodiments, crossfade between an equalized signal and an unprocessed signal may be used based on a position of the sound source between the center of the user's head and the surface of the user's head.

[0080] In some embodiments, the EQ filter may be automatically computed as an average of the filters used to render a source on a surface of a head of the user. The EQ filter may be exposed to the user as a set of tunable/configurable parameters. In some embodiments, the tunable/configurable parameters may include control frequencies and associated gains.

[0081] FIG. 12 illustrates a signal flow 1200 that may be implemented to render a sound source in a far-field, according to some embodiments. As illustrated in FIG. 12, a far-field distance attenuation 1220 can be applied to an input signal 1210, such as described above. A common EQ filter 1230 (e.g., a source radiation filter) may be applied to the result to model sound source radiation; the output of the filter 1230 can be split and sent to separate left and right channels, with delay (1240A/1240B) and VSA (1250A/1250B) functions applied to each channel, such as described above with respect to FIG. 5, to result in left ear and right ear signals 1290A/1290B.

[0082] FIG. 13 illustrates a signal flow 1300 that may be implemented to render a sound source in a near-field, according to some embodiments. As illustrated in FIG. 13, a far-field distance attenuation 1320 can be applied to an input signal 1310, such as described above. The output can be split into left/right channels, and separate EQ filters may be applied to each ear (e.g., left ear near-field and source radiation filter 1330A for a left ear, and right ear near-field and source radiation filter 1330B for a right ear) to model sound source radiation as well as nearfield ILD effects, such as described above. The filters can be implemented as one for each ear, after the left and right ear signals have been separated. Note that in this case, any other EQ applied to both ears could be folded into those filters (e.g., the left ear near-field and source radiation filter and the right ear near-field and source radiation filter) to avoid additional processing. Delay (1340A/1340B) and VSA (1350A/1350B) functions can then be applied to each channel, such as described above with respect to FIG. 5, to result in left ear and right ear signals 1390A/1390B.

[0083] In some embodiments, to optimize computing resources, a system may automatically switch between the signal flows 1200 and 1300, for example, based on whether the sound source to be rendered is in the far-field or in the near-field. In some embodiments, a filter state may need to be copied between the filters (e.g., the source radiation filter, the left ear near-field and source radiation filter and the right ear near-field and source radiation filter) during transitioning in order to avoid processing artifacts.

[0084] In some embodiments, the EQ filters described above may be bypassed when their settings are perceptually equivalent to a flat magnitude response with 0dB gain. If the

response is flat but with a gain different than zero, a broadband gain may be used to efficiently achieve the desired result.

[0085] FIG. 14 illustrates a signal flow 1400 that may be implemented to render a sound source in a near-field, according to some embodiments. As illustrated in FIG. 14, a far-field distance attenuation 1420 can be applied to an input signal 1410, such as described above. A left ear near-field and source radiation filter 1430 can be applied to the output. The output of 1430 can be split into left/right channels, and a second filter 1440 (e.g., a right-left ear near-field and source radiation difference filter) can then be used to process the right ear signal. The second filter models a difference between right and left ear nearfield and source radiation effects. In some embodiments, a difference filter may be applied to the left ear signal. In some embodiments, a difference filter may be applied to a contralateral ear, which may depend on a position of the sound source. Delay (1450A/1450B) and VSA (1460A/1460B) functions can be applied to each channel, such as described above with respect to FIG. 5, to result in left ear and right ear signals 1490A/1490B.

[0086] A head coordinate system may be used for computing acoustic propagation from an audio object to ears of a listener. A device coordinate system may be used by a tracking device (such as one or more sensors of a wearable head device in an augmented reality system, such as described above) to track position and orientation of a head of a listener. In some embodiments, the head coordinate system and the device coordinate system may be different. A center of the head of the listener may be used as the origin of the head coordinate system, and may be used to reference a position of the audio object relative to the listener with a forward direction of the head coordinate system defined as going from the center of the head of the listener to a horizon in front of the listener. In some embodiments, an arbitrary point in space may be used as the origin of the device coordinate system. In some embodiments, the origin of the device coordinate system may be a point located in between optical lenses of a visual projection system of the tracking device. In some embodiments, the forward direction of the device coordinate system may be referenced to the tracking device itself, and dependent on the position of the tracking device on the head of the listener. In some embodiments, the tracking device may have a non-zero pitch (i.e. be tilted up or down) relative to a horizontal plane of the head coordinate system, leading to a misalignment between the forward direction of the head coordinate system and the forward direction of the device coordinate system.

[0087] In some embodiments, the difference between the head coordinate system and the device coordinate system may be compensated for by applying a transformation to the position of the audio object relative to the head of the listener. In some embodiments, the difference in the origin of the head coordinate system and the device coordinate system may be compensated for by translating the position of the audio objects relative to the head of the listener by an amount equal to the distance between the origin of the head coordinate system and the origin of the device coordinate system reference points in three dimensions (e.g., x, y, and z). In some embodiments, the difference in angles between the head coordinate system axes and the device coordinate system axes may be compensated for by applying a rotation to the position of the audio object relative to the head of the

listener. For instance, if the tracking device is tilted downward by N degrees, the position of the audio object could be rotated downward by N degrees prior to rendering the audio output for the listener. In some embodiments, audio object rotation compensation may be applied before audio object translation compensation. In some embodiments, compensations (e.g., rotation, translation, scaling, and the like) may be taken together in a single transformation including all the compensations (e.g., rotation, translation, scaling, and the like).

[0088] FIGS. 15A-15D illustrate examples of a head coordinate system 1500 corresponding to a user and a device coordinate system 1510 corresponding to a device 1512, such as a head-mounted augmented reality device as described above, according to embodiments. FIG. 15A illustrates a top view of an example where there is a frontal translation offset 1520 between the head coordinate system 1500 and the device coordinate system 1510. FIG. 15B illustrates a top view of an example where there is a frontal translation offset 1520 between the head coordinate system 1500 and the device coordinate system 1510, as well as a rotation 1530 around a vertical axis. FIG. 15C illustrates a side view of an example where there are both a frontal translation offset 1520 and a vertical translation offset 1522 between the head coordinate system 1500 and the device coordinate system 1510. FIG. 15D shows a side view of an example where there are both a frontal translation offset 1520 and a vertical translation offset 1522 between the head coordinate system 1500 and the device coordinate system 1510, as well as a rotation 1530 around a left/right horizontal axis.

[0089] In some embodiments, such as in those depicted in FIGS. 15A-15D, the system may compute the offset between the head coordinate system 1500 and the device coordinate system 1510 and compensate accordingly. The system may use sensor data, for example, eye-tracking data from one or more optical sensors, long term gravity data from one or more inertial measurement units, bending data from one or more bending/head-size sensors, and the like. Such data can be provided by one or more sensors of an augmented reality system, such as described above.

[0090] Various exemplary embodiments of the disclosure are described herein. Reference is made to these examples in a non-limiting sense. They are provided to illustrate more broadly applicable aspects of the disclosure. Various changes may be made to the disclosure described and equivalents may be substituted without departing from the true spirit and scope of the disclosure. In addition, many modifications may be made to adapt a particular situation, material, composition of matter, process, process act(s) or step(s) to the objective(s), spirit or scope of the present disclosure. Further, as will be appreciated by those with skill in the art that each of the individual variations described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present disclosure. All such modifications are intended to be within the scope of claims associated with this disclosure.

[0091] The disclosure includes methods that may be performed using the subject devices. The methods may include the act of providing such a suitable device. Such provision may be performed by the end user. In other words, the “providing” act merely requires the end user obtain, access,

approach, position, set-up, activate, power-up or otherwise act to provide the requisite device in the subject method. Methods recited herein may be carried out in any order of the recited events which is logically possible, as well as in the recited order of events.

[0092] Exemplary aspects of the disclosure, together with details regarding material selection and manufacture have been set forth above. As for other details of the present disclosure, these may be appreciated in connection with the above-referenced patents and publications as well as generally known or appreciated by those with skill in the art. The same may hold true with respect to method-based aspects of the disclosure in terms of additional acts as commonly or logically employed.

[0093] In addition, though the disclosure has been described in reference to several examples optionally incorporating various features, the disclosure is not to be limited to that which is described or indicated as contemplated with respect to each variation of the disclosure. Various changes may be made to the disclosure described and equivalents (whether recited herein or not included for the sake of some brevity) may be substituted without departing from the true spirit and scope of the disclosure. In addition, where a range of values is provided, it is understood that every intervening value, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the disclosure.

[0094] Also, it is contemplated that any optional feature of the variations described may be set forth and claimed independently, or in combination with any one or more of the features described herein. Reference to a singular item, includes the possibility that there are plural of the same items present. More specifically, as used herein and in claims associated hereto, the singular forms “a,” “an,” “said,” and “the” include plural referents unless the specifically stated otherwise. In other words, use of the articles allow for “at least one” of the subject item in the description above as well as claims associated with this disclosure. It is further noted that such claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitation.

[0095] Without the use of such exclusive terminology, the term “comprising” in claims associated with this disclosure shall allow for the inclusion of any additional element—irrespective of whether a given number of elements are enumerated in such claims, or the addition of a feature could be regarded as transforming the nature of an element set forth in such claims. Except as specifically defined herein, all technical and scientific terms used herein are to be given as broad a commonly understood meaning as possible while maintaining claim validity.

[0096] The breadth of the present disclosure is not to be limited to the examples provided and/or the subject specification, but rather only by the scope of claim language associated with this disclosure.

What is claimed is:

1. A method comprising:

determining a reference point based on a location of a wearable head device; and

for each of a respective left ear and right ear of the user: determining, for a virtual speaker array associated with a sphere concentric with the reference point, the

sphere having a first radius, a virtual speaker position substantially collinear with a source location and a position of the respective ear, wherein the determined virtual speaker position comprises a location substantially on a surface of the sphere and the source location corresponds to an audio signal;

determining a distance between the source location and the position of the respective ear;

determining whether the distance is greater than a minimum distance;

in accordance with a determination that the distance is greater than the minimum distance, determining a parameter according to the distance;

in accordance with a determination that the distance is not greater than the minimum distance, determining the parameter according to the minimum distance;

generating, based on the parameter and based further on the audio signal, an output audio signal for the respective ear; and

presenting the output audio signal to the respective ear of the user via one or more speakers.

2. The method of claim 1, wherein the source location is separated from the reference point by a distance less than the first radius.

3. The method of claim 1, wherein the source location is separated from the reference point by a distance greater than the first radius.

4. The method of claim 1, wherein the source location is separated from the reference point by a distance equal to the first radius.

5. The method of claim 1, wherein said generating the output audio signal comprises applying an interaural time difference to the audio signal.

6. The method of claim 1, wherein the output audio signal is generated further based on a transfer function.

7. The method of claim 6, wherein the transfer function is selected from a plurality of transfer functions, wherein each transfer function of the plurality of transfer functions describes a relationship between a listener and an audio source, the audio source separated from the listener by a distance equal to the first radius.

8. The method of claim 1, wherein the wearable head device comprises the one or more speakers.

9. The method of claim 1, further comprising identifying the source location corresponding to the audio signal.

10. The method of claim 1, wherein the parameter comprises one or more of gain, azimuth angle, elevation angle, and interaural time difference.

11. The method of claim 1, wherein the minimum distance is determined based on a signal level of an audio source at the source location.

12. A system comprising:

- a wearable head device;
- one or more speakers; and
- one or more processors configured to perform a method comprising:
 - determining a reference point based on a location of the wearable head device; and
 - for each of a respective left ear and right ear of the user:
 - determining, for a virtual speaker array associated with a sphere concentric with the reference point, the sphere having a first radius, a virtual speaker position substantially collinear with a source location and a position of the respective ear, wherein the determined virtual speaker position comprises a location substantially on a surface of the sphere and the source location corresponds to an audio signal;

- determining a distance between the source location and the position of the respective ear;
 - determining whether the distance is greater than a minimum distance;
 - in accordance with a determination that the distance is greater than the minimum distance, determining a parameter according to the distance;
 - in accordance with a determination that the distance is not greater than the minimum distance, determining the parameter according to the minimum distance;
 - generating, based on the parameter and based further on the audio signal, an output audio signal for the respective ear; and
 - presenting the output audio signal to the respective ear of the user via the one or more speakers.

13. The system of claim 12, wherein the source location is separated from the reference point by a distance equal to the first radius.

14. The system of claim 12, wherein said generating the output audio signal comprises applying an interaural time difference to the audio signal.

15. The system of claim 12, wherein the output audio signal is generated further based on a transfer function.

16. The system of claim 12, wherein the wearable head device comprises the one or more speakers.

17. The system of claim 12, the method further comprising identifying the source location corresponding to the audio signal.

18. The system of claim 12, wherein the parameter comprises one or more of gain, azimuth angle, elevation angle, and interaural time difference.

19. The system of claim 12, wherein the minimum distance is determined based on a signal level of an audio source at the source location.

20. A non-transitory computer-readable medium storing instructions which, when executed by one or more processors, cause the one or more processors to perform a method comprising:

- determining a reference point based on a location of a wearable head device; and
- for each of a respective left ear and right ear of the user:
 - determining, for a virtual speaker array associated with a sphere concentric with the reference point, the sphere having a first radius, a virtual speaker position substantially collinear with a source location and a position of the respective ear, wherein the determined virtual speaker position comprises a location substantially on a surface of the sphere and the source location corresponds to an audio signal;
 - determining a distance between the source location and the position of the respective ear;
 - determining whether the distance is greater than a minimum distance;
 - in accordance with a determination that the distance is greater than the minimum distance, determining a parameter according to the distance;

in accordance with a determination that the distance is not greater than the minimum distance, determining the parameter according to the minimum distance;

generating, based on the parameter and based further on the audio signal, an output audio signal for the respective ear; and

presenting the output audio signal to the respective ear of the user via one or more speakers.

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